

 <i>Ministero della Salute</i> Direzione generale della ricerca e dell'innovazione in sanità PNRR: M6/C2_CALL 2023 Full Proposal	 Finanziato dall'Unione europea NextGenerationEU
Project Code: PNRR-MCNT2-2023-12377755	Call section: Malattie Croniche non Trasmissibili (MCnT2) ad alto impatto sui sistemi sanitari
Applicant Institution: Ospedale San Raffaele - Milano	Applicant/PI Coordinator: CHITI ARTURO

1 - General information

Project code: PNRR-MCNT2-2023-12377755	Project topic: E1) Malattie croniche non trasmissibili (MCnT2) ad alto impatto sui sistemi sanitari e socio-assistenziali: innovazioni in campo diagnostico
PI / Coordinator: CHITI ARTURO	Applicant Institution: Ospedale San Raffaele - Milano

Call section: Malattie Croniche non Trasmissibili (MCnT2) ad alto impatto sui sistemi sanitari e socio-assistenziali

Proposal title: Advancing Lung Cancer Screening: Artificial Intelligence, Multimodal Imaging and Cutting-Edge Technologies for Early Detection and Characterization

Duration in months: 24

MDC primary: Diagnostica

MDC secondary: Oncologia

Project Classification IRG: Surgical Sciences, Biomedical Imaging, and Bioengineering

Project Classification SS: Medical Imaging - MEDI

Project Keyword 1: Applications of imaging systems and modification of diagnostic methods for use in: screening; characterizing physiological effects,, and assessing risk.

Project Request: **Animals:** ☐ **Humans:** ☒ **Clinical trial:** ☒

Project total financing request to the MOH: € 1.000.000

Free keywords: Screening; lung cancer; oncology, Artificial Intelligence; FDG; PET/CT; CT

Declarations

In case of a Synergy grant application 'Principal Investigator'(PI) means 'corresponding Principal Investigator on behalf of all Principal Investigators', and 'Host Institution' means 'corresponding Host Institution'.

1) The Principal Investigator declares to have the written consent of all participants on their participation and on the content of this proposal, as well as of any researcher mentioned in the proposal as participating in the project (either as other PI, team member or collaborator).	☒
2) The Principal Investigator declares that the information contained in this proposal is correct and complete.	☒
3) The Principal Investigator declares that all parts of this proposal comply with ethical principles (including the highest standards of research integrity — as set out, for instance, in the European Code of Conduct for Research Integrity — and including, in particular, avoiding fabrication, falsification, plagiarism or other research misconduct).	☒
4) The Principal Investigator is only responsible for the correctness of the information relating to his/her own organisation. Each applicant remains responsible for the correctness of the information related to him and declared above.	☒

Personal data protection

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The assessment of your grant application will involve the collection and processing of personal data (such as your name, address and CV), which will be performed pursuant to Regulation (EC) No 45/2001 on the protection of individuals with regard to the processing of personal data by the Community institutions and bodies and on the free movement of such data. Unless indicated otherwise, your replies to the questions in this form and any personal data requested are required to assess your grant application in accordance with the specifications of the call for proposals and will be processed solely for that purpose. Details concerning the purposes and means of the processing of your personal data as well as information on how to exercise your rights are available in the privacy statement. Applicants may lodge a complaint about the processing of their personal data with the European Data Protection Supervisor at any time.

Abstract

Lung cancer is the leading cause of cancer-related morbidity and mortality worldwide. The main strategies to reduce lung cancer-related deaths are smoking cessation and early detection by screening the asymptomatic high-risk population with low-dose computed tomography (LDCT). Widespread implementation of lung cancer screening, though, is currently hampered by several challenges, mainly related to correct interpretation of images, radiation exposure and high costs. Positron Emission Tomography (PET) with 2-[fluorine-18]-fluoro-2-deoxy-d-glucose ([18F]FDG) is a key player in clinical staging in several malignancies, including lung cancer. [18F]FDG PET/CT is a highly accurate non-invasive imaging modality in the differential diagnosis of solitary pulmonary nodules indeterminate on CT. In addition, whole-body PET/CT images are able to detect other incidental malignancies with high sensitivity, especially in selected high-risk patient populations. Deep learning is emerging as a powerful tool for the detection and characterisation of lung nodules on CT and [18F]FDG PET/CT imaging. We hypothesised that a comprehensive approach using multimodality imaging with LDCT and low dose [18F]FDG PET/CT integrated with artificial intelligence (AI) models may enable early detection of suspicious lung nodules in the context of a lung cancer screening programme in the high-risk population. Our main objectives will be: assessment of the potential added value of low dose [18F]FDG PET/CT in the early detection of lung cancer in the screening work-up of the high-risk population (cohort 1) (1); assessment of the role of LDCT alone in a control high-risk population (cohort 2) (2); development of a multimodal deep learning model by combining patient-specific data with morphological and metabolic features derived from imaging to predict lung cancer risk (3). This comprehensive multimodal approach will define a tailored screening programme for each individual patient, anticipating the correct diagnosis, reducing radiation exposure and costs to the healthcare system.

In order to best review your application, do you agree that the above non-confidential proposal title and abstract can be used, without disclosing your identity, when contacting potential reviewers?

Yes

2 - Participants & contacts

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Operative Units					
Institution that perform as UO	CF Institution	Department / Division / Laboratory	Role in the project	Southern Italy	SSN
1 - Ospedale San Raffaele - Milano	7636600962	Nuclear medicine department; thoracic surgery department; oncology department	Coordinator, recruitment of patients for LC screening, imaging acquisition and interpretation		X
2 - Ospedale San Giovanni di Dio e Ruggi d'Aragona	95044230654	Diagnostic imaging department	Recruitment of patients for LC screening, LDCT imaging acquisition and interpretation	X	X
3 - Campus Biomedico - Roma	97087620585	Department of Engineering	Development of the AI model to predict the risk of lung cancer based on clinical and imaging data		
4 - DIMES Università della Calabria	80003950781	Department of Electronic and Computer Bioengineering	Development of the AI model to predict the risk of lung cancer based on clinical and imaging data	X	

Principal Research Collaborators		
Key Personnel Name	Operative Unit	Role in the project
1 - Soda Paolo	Campus Biomedico - Roma	Responsible UO3 - Full Professor in Computer Science and Computer Engineering
2 - VERONESI GIULIA	Ospedale San Raffaele - Milano	Further collaborator UO1 - Director of the Program Robotic Thoracic Surgery and Professor of Thoracic Surgery Università Vita-Salute San Raffaele
3 - VELTRI PIERANGELO	DIMES Università della Calabria	Coordinator UO4 - Full Professor of Electronic and Computer Bioengineering
4 - Trubitsyna Irina	DIMES Università della Calabria	Further collaborator UO4 - RTD-B in Electronic and Computer Bioengineering
5 - Guarrasi Valerio	Campus Biomedico - Roma	Further collaborator UO3 - Postdoc at Università Campus Bio-Medico di Roma
6 Under 40 - Cuocolo Renato	Ospedale San Giovanni di Dio e Ruggi d'Aragona	Responsible UO2 - Associate Professor at Università degli studi di Salerno
7 Under 40 - Ferrara Roberto	Ospedale San Raffaele - Milano	Further collaborator UO1 - Medical oncologist and University Researcher at Università Vita-Salute San Raffaele

Key Personnel Name	Co-PI	Resp. CE	Resp. Animal	Birth Date	Gender
1 - Soda Paolo				19/01/1981	M
2 - VERONESI GIULIA				18/08/1969	F
3 - VELTRI PIERANGELO				30/04/1973	M
4 - Trubitsyna Irina				11/01/1978	F
5 - Guarrasi Valerio				21/04/1995	M
6 Under 40 - Cuocolo Renato				21/04/1986	M
7 Under 40 - Ferrara Roberto	X			10/06/1986	M

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Responsible who requests CE authorization: CHITI ARTURO

Additional research collaborators under 40 to hire

Key Personnel Name	Operative Unit	Birth Date	Gender	Role in the project	Degree	Actual Pos. and Inst.
0 - DE GIORGI MARCO	Ospedale San Giovanni di Dio e Ruggi d'Aragona	12/09/1992	M	Radiologist	Laurea magistrale in Medicine and surgery, Residency in diagnostic imaging	Radiology consultant at P.O. Umberto I Nocera Inferiore
1 - PONSIGLIONE ANDREA	Ospedale San Giovanni di Dio e Ruggi d'Aragona	23/03/1988	M	Radiologist	Laurea magistrale in Medicine and surgery, Residency in diagnostic imaging, PhD in Biomorphological and Surgical Sciences	Post-doctoral research fellow in diagnostic imaging at University of Naples Federico II
2 - PERILLO TERESA	Ospedale San Giovanni di Dio e Ruggi d'Aragona	31/01/1992	F	Radiologist	Laurea magistrale in Medicine and surgery, Residency in diagnostic imaging	Neuroradiology consultant at P.O. Umberto I Nocera Inferiore

2.1 Administrative data of participating

Operative Unit Number 1:

Address: Via Olgettina n. 60, 20132, Milano

PEC: dir.scientifica@hsr.postecert.it

Operative Unit Number 2:

Address: Largo Città di Ippocrate, 84131, Salerno

PEC: info@pec.sangiovannieruggi.it

Operative Unit Number 3:

Address: Via Álvaro del Portillo 21, 00128, Roma

PEC: dirgenuni@postasicura.unicampus.it

Operative Unit Number 4:

Address: Ponte Pietro Bucci, 87036, Quattromiglia (CS)

PEC: amministrazione@pec.unical.it

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Operative Unit Number 5 (self financing):

Address: -
PEC: -

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2.2 Principal Investigator (PI) Profile

Last Name: CHITI	Last name at birth:
First Name: ARTURO	Gender: M
Title: Principal investigator	Country of residence: ITALY
Nationality: Italiana	Country of Birth: ITALY
Date of birth: 17/05/1962	Place of Birth: Bologna
Official H index (Scopus or Web of Science): 51.0	
Scopus Author Id: 22955689900	ORCID ID: 0000-0002-5806-1856 RESEARCH ID: K-6524-2016
Contact address	
Current organisation name: Ospedale San Raffaele - Milano	
Current Department / Faculty / Institute / Laboratory name: Nuclear medicine department; thoracic surgery department; oncology department	
Street: IRCCS San Raffaele, Via Olgettina 60	
Postcode / Cedex: 20132	Town: Milano
Phone: +393200697132	Phone 2: +390282248524

Education / training				
Educational institution and location	Degree	Field of study	From year	To year
Università degli Studi di Milano	Specialization / Specializzazione	Nuclear Medicine	1989	1993
Università degli Studi di Milano	Single-cycle master's degree / Laurea magistrale a ciclo unico	Medicine and Surgery	1982	1987

Personal Statement:

Arturo Chiti is the director of the Department of Nuclear Medicine at IRCCS Ospedale San Raffaele and Full Professor of Diagnostic Imaging and Radiotherapy at Università Vita-Salute San Raffaele in Milan. He has extensive experience in conducting translational research projects using artificial intelligence techniques in oncology, particularly in lung cancer. His role will be to coordinate the study and supervise all research activities in UO1 and other UOs involved in the project. In addition, he will interpret FDG PET/CT images.

Positions and honors

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Positions					
Institution	Division / Research group	Location	Position	From year	To year
Università Vita-Salute San Raffaele	Diagnostic imaging and Radiotherapy	Milano	Full Professor	2023	0
IRCCS Ospedale San Raffaele	Department of Nuclear Medicine	Milano	Director	2023	0
Humanitas University	Nuclear Medicine Department	Pieve Emanuele (MI)	Director of the training program in Nuclear Medicine	2018	2023
Humanitas University	Department of Biomedical Sciences	Pieve Emanuele (MI)	Director	2018	2023
Humanitas University	Diagnostic Imaging	Pieve Emanuele (MI)	Full Professor	2015	2023
IRCCS Humanitas Research Hospital	Nuclear Medicine Department	Rozzano (MI)	Director	2000	2023
Humanitas University	Diagnostic Imaging department	Pieve Emanuele (MI)	Director of the training program in radiology	2016	2018
European Association of Nuclear Medicine	European School of the European Association of Nuclear Medicine	Wien (Austria)	Lecturer	2004	2012
Fondazione IRCCS Istituto Nazionale dei Tumori	Nuclear Medicine department	Milano	Associated Chief	1993	2000
Università degli Studi di Milano	Diagnostic Imaging	Milano	Tutor for the diagnostic imaging course	1992	1999

Other awards and honors

Winner of the Bianca Maria Doniselli Colombo award from the Soroptimist Club Varese (1997); Honorary Membership of European Society for Therapeutic Radiology and Oncology (ESTRO) (2011); winner of the Hevesy Gyorgy award from the Hungarian Society of Nuclear Medicine (MONT) (2013); winner of the Tom Miller Memorial Lecture Award from the Society of Nuclear Medicine and Molecular Imaging (2013).

Other CV informations

2013-2014 President Elect of the European Association of Nuclear Medicine (EANM); 2015-2016 President of the European Association of Nuclear Medicine (EANM). From 2019 to present Editor in Chief of the European Journal of Nuclear Medicine and Molecular Imaging; from 2015 to present Associate Editor of the European Journal of Nuclear Medicine and Molecular Imaging Research; 2017-2018 Editor in Chief of the European Journal of Hybrid Imaging.

Selected peer-reviewed publications of the PI valid for minimum expertise level								
Title	Type	Pag	Vol	Year	DOI	PMID	Cit.**	P.*
The Role of PET Imaging in the Differential Diagnosis between Radiation Necrosis and Recurrent Disease in Irradiated Adult-Type Diffuse Gliomas: A Systematic Review	Review	NOT_FO UND	15	2023	10.3390/cancers15020364	36672314	0	L
The Development of an Intelligent Agent to Detect and Non-Invasively Characterize Lung Lesions on CT Scans: Ready for the Real World?	Article	NOT_FO UND	15	2023	10.3390/cancers15020357	36672306	0	L

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Title	Type	Pag	Vol	Year	DOI	PMID	Cit.**	P.*
Image Embeddings Extracted from CNNs Outperform Other Transfer Learning Approaches in Classification of Chest Radiographs	Article	NOT_FOUND	12	2022	10.3390/diagnostics12092084	NOT_FOUND	1	L
Recurrence-specific supervised graph clustering for subtyping Hodgkin Lymphoma radiomic phenotypes	Article	2155-2158	2021	2021	10.1109/EMBC46164.2021.9629625	34891715	0	L
Preoperative [11C]methionine PET to personalize treatment decisions in patients with lower-grade gliomas	Article	1546-1556	24	2022	10.1093/neuonc/noac040	35171292	3	L
State-of-the-art of FAPI-PET imaging: a systematic review and meta-analysis	Review	4396-4414	48	2021	10.1007/s00259-021-05475-0	34173007	39	L
Radiomics and gene expression profile to characterise the disease and predict outcome in patients with lung cancer	Article	3643-3655	48	2021	10.1007/s00259-021-05371-7	33959797	25	L
Distributed learning: a reliable privacy-preserving strategy to change multicenter collaborations using AI	Review	3791-3804	48	2021	10.1007/s00259-021-05339-7	33847779	9	L
Imaging-based prediction of molecular therapy targets in NSCLC by radiogenomics and AI approaches: A systematic review	Review	NOT_FOUND	10	2020	10.3390/diagnostics10060359	NOT_FOUND	27	L
Methodological framework for radiomics applications in Hodgkin's lymphoma	Article	1	4	2020	10.1186/s41824-020-00078-8	NOT_FOUND	6	L
Computed tomography (CT)-derived radiomic features differentiate prevascular mediastinum masses as thymic neoplasms versus lymphomas	Article	951-960	125	2020	10.1007/s11547-020-01188-w	32306201	45	L
Imaging-based prediction of molecular therapy targets in NSCLC by radiogenomics and AI approaches: A systematic review	Review	NOT_FOUND	10	2020	10.3390/diagnostics10060359	NOT_FOUND	27	L
Convolutional neural networks promising in lung cancer T-parameter assessment on baseline FDG-PET/CT	Article	1	2018	2018	10.1155/2018/1382309	30510492	36	L
[¹⁸ F]FDG PET/CT features for the molecular characterization of primary breast tumors	Article	1945-1954	44	2017	10.1007/s00259-017-3770-9	28711994	44	C
Development of standardized image interpretation for 68Ga-PSMA PET/CT to detect prostate cancer recurrent lesions	Article	1622-1635	44	2017	10.1007/s00259-017-3725-1	28536833	40	L
FDG PET/CT as theranostic imaging in diagnosis of non-small cell lung cancer	Article	1713-1723	22	2017	10.2741/4567	28410141	10	L
Correlation of metabolic information on FDG-PET with tissue expression of immune markers in patients with non-small cell lung cancer (NSCLC) who are candidates for upfront surgery	Article	1954-1961	43	2016	10.1007/s00259-016-3425-2	27251642	89	L
PET/CT with [¹¹ C]-choline for evaluation of prostate cancer patients with biochemical recurrence: meta-analysis and critical review of available data	Review	55-69	43	2016	10.1007/s00259-015-3202-7	26450693	131	L

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Title	Type	Pag	Vol	Year	DOI	PMID	Cit.**	P.*
Quantitative analyses at baseline and interim PET evaluation for response assessment and outcome definition in patients with malignant pleural mesothelioma	Article	667-675	42	2015	10.1007/s00259-014-2960-y	25403555	31	L

* Position: F=First L=Last C=Correspondent O=Other N=Not applicable

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Selected peer-reviewed publications of the PI for the evaluation CV								
Title	Type	Pag	Vol	Year	DOI	PMID	Cit.**	
Towards clinical application of image mining: a systematic review on artificial intelligence and radiomics	Review	2656-2672	46	2019	10.1007/s00259-019-04372-x	31214791	112	
Prediction of disease-free survival by the PET/CT radiomic signature in non-small cell lung cancer patients undergoing surgery	Article	207-217	45	2018	10.1007/s00259-017-3837-7	28944403	114	
Texture analysis and machine learning to characterize suspected thyroid nodules and differentiated thyroid cancer: Where do we stand?	Review	1-8	99	2018	10.1016/j.ejrad.2017.12.004	29362138	73	
Guideline for PET/CT imaging of neuroendocrine neoplasms with ⁶⁸ Ga-DOTA-conjugated somatostatin receptor targeting peptides and ¹⁸ F-DOPA	Article	1588-1601	44	2017	10.1007/s00259-017-3728-y	28547177	205	
PET Radiomics in NSCLC: State of the art and a proposal for harmonization of methodology	Article	358	7	2017	10.1038/s41598-017-00426-y	28336974	98	
Correlation of metabolic information on FDG-PET with tissue expression of immune markers in patients with non-small cell lung cancer (NSCLC) who are candidates for upfront surgery	Article	1954-1961	43	2016	10.1007/s00259-016-3425-2	27251642	89	
PET/CT with ¹¹ C-choline for evaluation of prostate cancer patients with biochemical recurrence: meta-analysis and critical review of available data	Review	55-69	43	2016	10.1007/s00259-015-3202-7	26450693	131	
Long-Term Follow-up Results of the DANTE Trial, a Randomized Study of Lung Cancer Screening with Spiral Computed Tomography	Article	1166-1175	191	2015	10.1164/rccm.201408-1475OC	25760561	254	
FDG PET/CT: EANM procedure guidelines for tumour imaging: version 2.0	Review	328-354	42	2015	10.1007/s00259-014-2961-x	25452219	1497	
FDG PET and PET/CT: EANM procedure guidelines for tumour PET imaging: Version 1.0	Review	181-200	37	2010	10.1007/s00259-009-1297-4	19915839	846	

** Autocertificated

Grant						
Funded by Institution	Researcher inst. where grant is/was performed	Year	Title	Position in Projects	Fund (euro)	Source website grant listed

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Funded by Institution	Researcher inst. where grant is/was performed	Year	Title	Position in Projects	Fund (euro)	Source website grant listed
AIRC (Italian Association Against Cancer) - Investigator Grant	Humanitas University	2019	Image Mining and ctDNA to Improve Risk Stratification and Outcome Prediction in NSCLC applying Artificial Intelligence,	Coordinator	1.035.000,00	https://www.airc.it/fondazione/cosa-facciamo/cosa-finanziamo/investigat-or-grant
AIRC (Italian Association Against Cancer)	IRCCS Humanitas Research Hospital	2019	Radiomic features of tumor and of liver tumor interface in patients with colorectal liver metastases	Collaborator	540.000,00	https://programmi5perimille.airc.it/wp-content/uploads/2021/10/Rendicontazione-contributo-5xmille-2018-Ministero-DellUniversita-e-della-Ricerca-Allegato-B.pdf
LILT (Italian League to Fight Cancer)	IRCCS Humanitas Research Hospital	2018	Caratterizzazione non invasiva mediante tecniche di intelligenza artificiale delle lesioni polmonari evidenziate alla TC del torace	Collaborator	80.000,00	NA
ERA-NET TRANSCAN-2 funded by the European Commission under the EU framework programme Horizon2020 - 4th Joint Transnational Call (JTC)	IRCCS Humanitas Research Hospital	2017	Profiling radioRESistant Differentiated thyroid Cancer: genes, immunity, cancer stem cells and epithelial-mesenchymal transition (PREDICT)	Collaborator	765.662,00	https://www.era-learn.eu/network-information/networks/transcan-2/4th-joint-transnational-call-jtc-2017/profiling-radioresistant-differentiated-thyroid-cancer-genes-immunity-cancer-stem-cells-and-epithelial-mesenchymal-transition
AIRC (Italian Association Against Cancer) - Investigator Grant	Humanitas University	2016	Radiomics of FDG PET and CT to assess genetic profile and outcome in patients with lung lesions	Coordinator	240.000,00	https://programmi5perimille.airc.it/wp-content/uploads/2021/03/All-B-_Elenco-dei-progetti-finanziati.pdf
Bando Conto Capitale 2010 (Italian Ministry of Health)	IRCCS Humanitas Research Hospital	2010	Funding for a PET-CT at the IRCCS Humanitas Research Hospital	Coordinator	2.800.000,00	NA
AIRC	IRCCS Humanitas Research Hospital	2008	NA	Coordinator	120.000,00	NA

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2.3 CO-PI Profile

Last Name: Ferrara	Last name at birth:
First Name: Roberto	Gender: M
Title: Further collaborator UO1 - Medical oncologist and University Researcher at Università Vita-Salute San Raffaele	Country of residence: ITALY
Nationality: Italiana	Country of Birth: ITALY
Date of birth: 10/06/1986	Place of Birth: Avellino
Official H index (Scopus or Web of Science): 28.0	
Scopus Author Id: 56680868500	ORCID ID: 0000-0001-6443-5642 RESEARCH ID: AAC-2806-2019
Contact address	
Current organisation name: Ospedale San Raffaele - Milano	
Current Department / Faculty / Institute / Laboratory name: Nuclear medicine department; thoracic surgery department; oncology department	
Street: Università Vita e Salute San Raffaele	
Postcode / Cedex: 20132	Town: Milano
Phone: +393274737878	Phone 2:

Education / training				
Educational institution and location	Degree	Field of study	From year	To year
Molecular Immunology Unit and Thoracic Oncology Unit at IRCCS Istituto Nazionale dei Tumori (Open University Program)	PhD	Characterization of hyperprogressive disease biomarkers in lung cancer patients and in vivo models. PhD project title: "HYPERBOLIC: HYPER-progression Biological Outlines in advanced Lung cancer upon Immune Checkpoints blockade"	2018	2020
Università degli studi di Verona	Specialization / Specializzazione	Medical oncology	2012	2017
Università degli studi di Napoli Federico II	Single-cycle master's degree / Laurea magistrale a ciclo unico	Medicine and surgery	2009	2012

Personal Statement:

Roberto Ferrara is a clinical oncologist at IRCCS Ospedale San Raffaele and a researcher at Università Vita-Salute San Raffaele in Milan. He is conducting several research projects in patients with lung cancer, particularly in the field of immunotherapy. His role will be to collaborate with nuclear medicine physicians and thoracic surgeons in the management of patients following screening procedures.

Positions and honors

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Project Code: PNRR-MCNT2-2023-12377755	Call section: Malattie Croniche non Trasmissibili (MCnT2) ad alto impatto sui sistemi sanitari
Applicant Institution: Ospedale San Raffaele - Milano	Applicant/PI Coordinator: CHITI ARTURO

Positions					
Institution	Division / Research group	Location	Position	From year	To year
IRCSS Istituto Nazionale dei Tumori	Oncology department	Milan	Medical Oncologist	2018	2022
Institut Gustave Roussy	Laboratory of Immune-Oncology	Villejuif (Paris)	Translational Research Fellow	2017	2018
Institut Gustave Roussy	Thoracic Oncology Unit	Villejuif (Paris)	Clinical Research Fellow	2016	2017

Other awards and honors

IASLC (International Association for the Study of Lung Cancer) Award 2017/2018; ESMO (European Society of Medical Oncology) Best Poster Award 2017; Winner of FONICAP (Forza Operativa Nazionale Interdisciplinare contro il Cancro del Polmone) Award 2017; ESMO Best Poster Award 2018; prize 2 Grandangolo in Oncologia 2018 Fondazione Federico Calabresi Onlus 2018; ELCC (European Lung Cancer Conference) Best Poster Award 2019; ASCO (American Society of Clinical Oncology) 2019 Merit Award 2019

Other CV informations

Participation to AIOM Guideline Committee and extension of AIOM Lung Cancer Guidelines 2020 and 2021; Participation to ESMO LGP with specific focus on ESMO policy, networking and educational activities; Member of ESMO Faculty Board on Non-Metastatic non-small cell lung cancer and other thoracic malignancies; Systematic reviews and metanalysis training Cochrane 2018; Diplôme Universitaire Européen de Recherche Transationnelle Et Clinique en Cancérologie DUERTECC.

Selected peer-reviewed publications of the Co-PI valid for minimum expertise level								
Title	Type	Pag	Vol	Year	DOI	PMID	Cit.**	P.*
Single or combined immune checkpoint inhibitors compared to first-line platinum-based chemotherapy with or without bevacizumab for people with advanced non-small cell lung cancer	Review	NOT_FOUND	2021	2021	10.1002/14651858.CD013257.pub3	33930176	13	F
Circulating T-cell immunosenescence in patients with advanced non-small cell lung cancer treated with single-agent PD-1/PD-L1 inhibitors or platinum-based chemotherapy	Comment	492-503	27	2021	10.1158/1078-0432.CCR-20-1420	32887723	44	F
Single or combined immune checkpoint inhibitors compared to first-line platinum-based chemotherapy with or without bevacizumab for people with advanced non-small cell lung cancer	Review	NOT_FOUND	2020	2020	10.1002/14651858.CD013257.pub2	33316104	5	F
Comparison of fast-progression, hyperprogressive disease, and early deaths in advanced non-small-cell lung cancer treated with PD-1/PD-L1 inhibitors or chemotherapy	Article	829-840	4	2020	10.1200/PO.20.00021	NOT_FOUND	15	F
Hyperprogressive Disease upon Immune Checkpoint Blockade: Focus on Non-small Cell Lung Cancer	Review	NOT_FOUND	22	2020	10.1007/s11912-020-00908-9	32296957	13	L

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Title	Type	Pag	Vol	Year	DOI	PMID	Cit.**	P.*
How to recognize and manage hyper-progression and pseudoprogression during immune checkpoint blockade in non-small cell lung cancer	Review	NOT_FOUND	2	2019	10.21037/pcm.2019.10.03	NOT_FOUND	2	L
Pseudoprogression in Non-Small Cell Lung Cancer upon Immunotherapy: Few Drops in the Ocean?	Editorial	328-331	14	2019	10.1016/j.jtho.2018.12.011	30782378	26	F
Single or combined immune checkpoint inhibitors compared to first-line chemotherapy with or without bevacizumab for people with advanced non-small cell lung cancer	Review	NOT_FOUND	2019	2019	10.1002/14651858.CD013257	NOT_FOUND	32	F
Choosing wisely first line immunotherapy in non-small cell lung cancer (NSCLC): what to add and what to leave out	Review	39-51	75	2019	10.1016/j.ctrv.2019.03.004	30954906	94	C
Clinical and Translational Implications of RET Rearrangements in Non-Small Cell Lung Cancer	Review	27-45	13	2018	10.1016/j.jtho.2017.10.021	29128428	131	F
Hyperprogressive Disease in Patients with Advanced Non-Small Cell Lung Cancer Treated with PD-1/PD-L1 Inhibitors or with Single-Agent Chemotherapy	Article	1543-1552	4	2018	10.1001/jamaoncol.2018.3676	30193240	449	F
Immunosenescence and immunecheckpoint inhibitors in non-small cell lung cancer patients: Does age really matter?	Review	60-68	60	2017	10.1016/j.ctrv.2017.08.003	28889085	100	C
Tubulin inhibitors in non-small cell lung cancer: looking back and forward	Review	1113-1129	17	2016	10.1517/14656566.2016.1157581	26898217	16	F

* Position: F=First L=Last C=Correspondent O=Other N=Not applicable

** Autocertificated

Grant						
Funded by Institution	Researcher inst. where grant is/was performed	Year	Title	Position in Projects	Fund (euro)	Source website grant listed
IASLC (International Association for the Study of Lung Cancer) Foundation Young Investigator Award 2019	Istituto Nazionale dei Tumori	2019	HYPERBOLIC: HYPER-progression Biological Outlines in advanced Lung cancer upon Immune Checkpoints blockade	Coordinator	50.000,00	https://www.iaslc.org/iaslc-foundation/research-grant-recipients

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Applicant Institution: Ospedale San Raffaele - Milano	Applicant/PI Coordinator: CHITI ARTURO

2.3 Research Collaborators n. 1

Last Name: Soda

First Name: Paolo

Last name at birth:

Gender: M

Title: Responsible UO3 - Full Professor in Computer Science and Computer Engineering

Country of residence: ITALY

Nationality: Italiana

Country of Birth: ITALY

Date of birth: 19/01/1981

Place of Birth: Roma

Official H index (Scopus or Web of Science): 24.0

Scopus Author Id:22036116900

ORCID ID:0000-0003-2621-072X

RESEARCH ID:K-8126-2016

Contact address

Current organisation name: Campus Biomedico - Roma

Current Department / Faculty / Institute / Laboratory name: Department of Engineering

Street: via alvaro del portillo 21

Postcode / Cedex: 00128

Town: Roma

Phone:+393335227124

Phone 2:

Education / training				
Educational institution and location	Degree	Field of study	From year	To year
Università Campus Bio-Medico di Roma	PhD	Biomedical Engineering	2004	2008
Università Campus Bio-Medico di Roma	Master's Degree / Laurea Magistrale	Biomedical Engineering	2002	2004
Università Campus Bio-Medico di Roma	Bachelor Degree / Laurea Triennale	Biomedical Engineering	1999	2002

Personal Statement:

Paolo Soda is Full Professor of Computer Science and Computer Engineering at Università Campus Bio-Medico di Roma and visiting Professor in AI and Biomedical Engineering at Umea University. His scientific interests include biomedical applications of classification techniques, signal, image and video processing and analysis. His role will be to coordinate the activities of UO3 and to collaborate with the other UOs involved in the project.

Positions and honors

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Applicant Institution: Ospedale San Raffaele - Milano	Applicant/PI Coordinator: CHITI ARTURO

Positions					
Institution	Division / Research group	Location	Position	From year	To year
Umea University	Department of Radiation Science	Umea, Sweden	Visiting Professot in AI and Biomedica Engineering	2022	0
Università Campus Bio-Medico di Roma	Department of Engineering	Roma	Research and Third Mission Coordinator of the Department of Engineering	2022	0
Università Campus Bio-Medico di Roma	Department of Engineering	Roma	Full Professor in Computer Science and Computer Engineering	2022	0
Università Campus Bio-Medico di Roma	Department of Engineering	Roma	Vice-Coordinator of the National PhD program in AI, area health and life sciences	2021	0
Università Campus Bio-Medico di Roma	Department of Engineering	Roma	Associate Professor in Computer Science and Computer Engineering	2015	2022
Università Campus Bio-Medico di Roma	Department of Engineering	Roma	Scientific Director of the Second Level Master in Digital Open Innovation & Entrepreneurship	2020	2023
Università Campus Bio-Medico di Roma in conjunction with Centro Diagnostico Italiano (CDI)	Department of Medicine	Roma	Co-Chair of the Collaborative Laboratory of Precision Medicine & BioData Analytics	2018	0
Università Campus Bio-Medico di Roma	Department of Medicine	Rome	Assistant Professor in Computer Science and Computer Engineering	2009	2014
Università Campus Bio-Medico di Roma	Department of Engineering	Roma	PostDoc	2008	2009

Other awards and honors

Best student paper award at IEEE Life Sciences Conference (2017); student travel award at IEEE Int Conf on Bioinformatics and Biomedicine (2018); best paper award at IEEE Int Conf on Cognitive Informatics & Cognitive Computing (2019); best paper award at IEEE Int Conf on Computer-Based Medical Systems (2021); winner of the competition "All against COVID-19: Screening X-ray Images for COVID-19 Infection", IEEE (2021); winner of the competition: "COVID CXR Hackathon", Expo Dubai (2022).

Grant						
Funded by Institution	Researcher inst. where grant is/was performed	Year	Title	Position in Projects	Fund (euro)	Source website grant listed
Ministero dell'Università e della Ricerca (MUR)	Consiglio Nazionale delle Ricerche (CNR), Roma	2023	FAIR: Future Artificial Intelligence Research	Collaborator	114.500.000,00	https://future-ai-research.it/
Ministero degli Affari Esteri e della Cooperazione Internazionale (MAECI)	Università Campus Bio-Medico di Roma	2023	Trustworthy AI-driven next generation precision medicine in COVID-19	Coordinator	150.000,00	NA
Ministero dell'università e della ricerca (MUR)	Consiglio Nazionale delle Ricerche (CNR), Roma	2022	Fit 4 Medical Robotics	Collaborator	160.400.000,00	NA



Ministero della Salute

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Project Code: PNRR-MCNT2-2023-12377755	Call section: Malattie Croniche non Trasmissibili (MCnT2) ad alto impatto sui sistemi sanitari
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Funded by Institution	Researcher inst. where grant is/was performed	Year	Title	Position in Projects	Fund (euro)	Source website grant listed
Ministero dello Sviluppo Economico (MISE)	Bracco Imaging	2022	Piattaforma per la Medicina di Precisione. Intelligenza Artificiale e Diagnostica Clinica Integrata.	Coordinator	150.000,00	NA
PNRR	Sapienza Università di Roma	2022	Rome Technopole - Flagship project 3	Coordinator	75.000.000,00	NA
Fondazione CRUI	Università Campus Bio-Medico di Roma	2021	Go for IT program	Coordinator	30.000,00	NA
Regione Lazio	Università Campus Bio-Medico di Roma	2021	We-ease-it: a smart and intelligent ambulatory for Hospital 4.0	Coordinator	75.000,00	NA
EU commission	Zilina University	2020	University-Industry Educational Centre in Advanced Biomedical and Medical Informatics	Coordinator	639.904,00	https://cebmi.fri.uniza.sk/
Fondazione Roma	Università Campus Bio-Medico di Roma	2016	A novel approach to identify COPD phenotypes, forecast clinical course and plan the therapeutic strategy	Coordinator	280.800,00	NA

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Applicant Institution: Ospedale San Raffaele - Milano	Applicant/PI Coordinator: CHITI ARTURO

2.4 Research Collaborators n. 2

Last Name: VERONESI First Name: GIULIA	Last name at birth: Gender: F
Title: Further collaborator UO1 - Director of the Program Robotic Thoracic Surgery and Professor of Thoracic Surgery Università Vita-Salute San Raffaele	Country of residence: ITALY Country of Birth: ITALY Place of Birth: Milano
Nationality: Italiana Date of birth: 18/08/1969	
Official H index (Scopus or Web of Science): 46.0	
Scopus Author Id: 15519681600	ORCID ID: 0000-0001-6595-7776 RESEARCH ID: AAA-6327-2021
Contact address	
Current organisation name: Ospedale San Raffaele - Milano	
Current Department / Faculty / Institute / Laboratory name: Nuclear medicine department; thoracic surgery department; oncology department	
Street: Via Olgettina 60	
Postcode / Cedex: 20132	Town: Milano
Phone: +393398404074	Phone 2:

Education / training				
Educational institution and location	Degree	Field of study	From year	To year
Università degli Studi di Milano	Specialization / Specializzazione	General and Thoracic Surgery	1995	2000
Università degli Studi di Milano	Single-cycle master's degree / Laurea magistrale a ciclo unico	Medicina e Chirurgia	1989	1994

Personal Statement:

Giulia Veronesi is the director of the Robotic Thoracic Surgery at the IRCCS Ospedale San Raffaele and Associate Professor at Università Vita-Salute San Raffaele of Milan. She is leading several research projects aimed at developing effective screening programmes to enable early diagnosis and treatment of lung cancer. Her role will be to select eligible patients for [18F]FDG PET/CT screening and to collaborate with nuclear medicine physicians and oncologists in the management of patients following screening procedures.

Positions and honors

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Positions					
Institution	Division / Research group	Location	Position	From year	To year
Università Vita-Salute San Raffaele	Thoracic Surgery	Milano	Associate Professor	2020	0
IRCCS Ospedale San Raffaele	Program Robotic Thoracic Surgery at the Department of Thoracic Surgery	Milano	Director	2020	0
IRCCS Humanitas Research Hospital	Robotic Surgery Unit at the Division of Thoracic and General Surgery	Rozzano (MI)	Director	2015	2019
European Institute of Oncology (IEO)	Unit of Lung Cancer Early Detection at the Division of Thoracic Surgery	Milano	Director	2010	2015
European Institute of Oncology (IEO)	Division of Thoracic Surgery	Milano	Deputy Director	2004	2009
European Institute of Oncology (IEO)	Division of Thoracic Surgery	Milano	Senior Assistant	2003	2004
European Institute of Oncology (IEO)	Division of Thoracic Surgery	Milano	Assistant Surgeon	2001	2002

Other awards and honors

NA

Grant						
Funded by Institution	Researcher inst. where grant is/was performed	Year	Title	Position in Projects	Fund (euro)	Source website grant listed
Centro Nazionale per la prevenzione e il controllo delle malattie (CCM)	Ospedale San Raffaele	2019	Progetto Pilota di un programma di screening per il tumore polmonare integrato con la cessazione del fumo: percorsi, selezione dei soggetti e protocolli diagnostici, in vista di una valutazione HTA	Collaborator	400.000,00	https://www.ccm-network.it/progetto.jsp?id=node/2046&idP=740
Bando PNRR - Proof-of-Concept	Ospedale San Raffaele	2022	A new portable device for PRE-hospital non-invasive VENTilatory support in acute respiratory failure (PREVENT)	Collaborator	980.000,00	https://www.salute.gov.it/imgs/C_17_bandi_295_1_file.pdf
Inail Grant Support (BRiC)	Università Vita-Salute San Raffaele	2019	Development of an artificial intelligence model in lung cancer screening for the diagnosis of lung nodules and risk stratification in subjects with occupational and / or smoking exposure	Collaborator	252.000,00	NA
AIRC (Italian Association Against Cancer)	Humanitas University and Università Vita-Salute San Raffaele	2017-2022	Circulating and imaging biomarkers to improve lung cancer management and early detection	Coordinator	765.000,00	https://www.airc.it/ricercatori/i-nostri-ricercatori/giulia-veronesi

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Funded by Institution	Researcher inst. where grant is/was performed	Year	Title	Position in Projects	Fund (euro)	Source website grant listed
Ministero della Salute - Bando di Ricerca Finalizzata	IRCCS Humanitas Research Hospital	2016	Optimised lung cancer screening to prevent cardiovascular and pulmonary diseases coupled with primary prevention	Coordinator	405.000,00	NA
ERA-NET TRANSCAN-2- Joint Transnational Call 2016 (JTC 2016)	IRCCS Humanitas Research Hospital	2016	Validation of multiparametric models and Circulating and imaging biomarkers to improve Lung cancer EARLY detection (CLEARLY)	Coordinator	1.630.749,00	https://www.transcanfp7.eu/index.php/abstract/clearly.html
Inail Grant Support	IRCCS Humanitas Research Hospital	2016-2018	Development of a surveillance protocol for early detection of lung cancer with LDTC for subjects exposed to asbestos	Coordinator	120.000,00	NA
LILT (Italian League for the fight against cancer)	IRCCS Humanitas Research Hospital	2015	Pilot study on the role of circulating tumor cells (CTC) in early detection of lung cancer	Coordinator	72.000,00	NA
Inail Grant support	European Institute of Oncology	2013-2014	Prevention and surveillance of thoracic diseases in Asbestos Exposed workers	Coordinator	140.000,00	NA
Ministero della Salute - Ricerca Finalizzata	European Institute of Oncology (IEO)	2007	Identification and isolation of normal and tumoral lung stem cells as a tool for the definition of new therapeutic strategies in lung cancer	Collaborator	390.000,00	https://www.salute.gov.it/portale/ricercaSanitaria/dettaglioContenutiRicercaSanitaria.jsp?lingua=italiano&id=5717&area=Ricerca%20sanitaria&menu=progetti20002008&tab=1

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2.5 Research Collaborators n. 3

Last Name: VELTRI
First Name: PIERANGELO

Last name at birth:

Gender: M

Title: Coordinator UO4 - Full Professor of Electronic and Computer Bioengineering

Country of residence: ITALY

Country of Birth: ITALY

Nationality: Italiana

Place of Birth: Cosenza

Date of birth: 30/04/1973

Official H index (Scopus or Web of Science): 26.0

Scopus Author Id: 15766584600

ORCID ID: 0000-0003-2494-0294

RESEARCH ID: NA

Contact address

Current organisation name: DIMES Università della Calabria

Current Department / Faculty / Institute / Laboratory name: Department of Electronic and Computer Bioengineering

Street: via P Bucci

Postcode / Cedex: 87036

Town: Rende

Phone: +393404898202

Phone 2: 3316718279

Education / training				
Educational institution and location	Degree	Field of study	From year	To year
Université Paris XI (Orsay)	PhD	Computer Science	1998	2002
Università della Calabria	Single-cycle master's degree / Laurea magistrale a ciclo unico	Computer Engineering	1991	1998

Personal Statement:

Pierangelo Veltri is Full Professor of Electronic and Computer Bioengineering at Università della Calabria. His research focuses on the application of deep learning techniques to the mining of electronic health records and the use of computational techniques for advanced analysis of radiology and nuclear medicine DICOM images. His role will be to coordinate the activities of UO4 and to collaborate with the other UOs involved in the project.

Positions and honors

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Applicant Institution: Ospedale San Raffaele - Milano	Applicant/PI Coordinator: CHITI ARTURO

Positions					
Institution	Division / Research group	Location	Position	From year	To year
Università della Calabria	DIMES Department of Computer, Modeling, Electronic and Systems Engineering - Electronic Bioengineering and Computer Science	Rende (Cosenza)	Full Professor	2022	0
Università Magna Graecia di Catanzaro	Department of Experimental and Clinical Medicine - Electronic and Computer Bioengineering	Catanzaro	Full Professor	2021	2022
Università Magna Graecia di Catanzaro	Department of Experimental and Clinical Medicine - Information Processing Systems	Catanzaro	Full Professor	2020	2021
Università Magna Graecia di Catanzaro	Department of Experimental and Clinical Medicine - Information Processing Systems	Catanzaro	Associate Professor	2012	2020
Università Magna Graecia di Catanzaro	Department of Experimental and Clinical Medicine - Information Processing Systems	Catanzaro	Researcher	2002	2012
Université Sorbonne-Paris-Nord (XIII) - Institut Galiee	Department of Computer Science	Villetaneuse (Paris)	Lecturer	200	2002
INRIA di Rocquencourt	Project VERSO	Paris	TMR Researcher	1998	2002

Other awards and honors

ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM BCB) best poster (2009); IBM Watson build challenge Italia 2018 with the project "VOTA - A Vocal tract analysis system"

Grant						
Funded by Institution	Researcher inst. where grant is/was performed	Year	Title	Position in Projects	Fund (euro)	Source website grant listed
Ministero dello Sviluppo Economico (MISE) - PON	Università Magna Graecia di Catanzaro	2019 - 2022	Validated Question Answering (VQA)	Coordinator	800.000,00	https://sites.google.com/unicz.it/vqa-project-pon-14-20/home-page
POR Regione Calabria	Università Magna Graecia di Catanzaro	2021-2023	Misurare Gestì il Modello innovativo socio-sanitariodelle Unità di Rete assistenziale per il Recepimento delle Esigenze, la Gestione Ed il Supporto Territoriale agli Inabili	Coordinator	103.800,14	https://www.webgenesys.it/-azienda/ricerca-sviluppo/

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Funded by Institution	Researcher inst. where grant is/was performed	Year	Title	Position in Projects	Fund (euro)	Source website grant listed
POR Regione Calabria	Università Magna Graecia di Catanzaro	2017 - 2019	Sistema IoT di Telemetria 4.0 per macchinari di movimentazione merci e terra, per applicazioni nell'ambito della logistica del magazzino, manifatturiera e dell'agroalimentare	Coordinator	75.838,36	NA
PIA	Università Magna Graecia di Catanzaro	2014- 2015	GIDAC - Sistema di annotazione bioimmagini radiologiche	Coordinator	107.400,00	NA
POR Regione Calabria	Università Magna Graecia di Catanzaro	2017 - 2019	SISTABENE: SiStema di Tracciabilità Avanzata per il BENEssere Alimentare	Collaborator	95.226,91	http://www.torredimezzo.com/sistabene/
POR Regione Calabria	Università Magna Graecia di Catanzaro	2017- 2019	Design e Sviluppo di un dispositivo biomedico innovativo per la diagnostica prevoce non invasiva del carcinoma prostatico (INNOPROST)	Collaborator	296.490,24	https://dmsc.unicz.it/progetti_ricerca_finanziati
POR Regione Calabria	Università Magna Graecia di Catanzaro	2017 - 2019	Nuovi sistemi di neuronavigazione in chirurgia maxillo facciale attraverso l'utilizzo di nuove placche endo-oralì come punti craniometrici	Collaborator	247.145,00	https://dmsc.unicz.it/progetti_ricerca_finanziati
Ministero dello Sviluppo Economico (MISE) - PON	Università Magna Graecia di Catanzaro	2013 - 2016	DICET - INMOTO - Organization of Cultural HEritage for Smart Tourism and Real-time Accessibility (OR.C.HE.S.T.R.A.)	Collaborator	1.668.205,60	http://www.ponrec.it/open-data/progetti/scheda-progetto?ProgettoID=5835
Ministero dello Sviluppo Economico (MISE) - PON	Università Magna Graecia di Catanzaro	2012 - 2015	SMART HEALTH CLUSTER OSDH - SMART FSE -STAYWELL	Collaborator	1.685.449,90	http://www.ponrec.it/open-data/progetti/scheda-progetto?ProgettoID=5833
Ministero dello Sviluppo Economico (MISE) - PON	Università Magna Graecia di Catanzaro	2009- 2012	TeCnologie Avanzate per l'innovazione e l'ottimizzazione dei pRocessi DIagnOstici, Terapeutici E di training dedicati alla gestione Clinica, interventistica e riabilitativa dei paziente affetti da sindromi coronariche acute.(CARDIOTECH)	Collaborator	237.753,50	http://www.ponrec.it/open-data/progetti/scheda-progetto?ProgettoID=5256

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Applicant Institution: Ospedale San Raffaele - Milano	Applicant/PI Coordinator: CHITI ARTURO

2.6 Research Collaborators n. 4

Last Name: Trubitsyna

First Name: Irina

Last name at birth:

Gender: F

Title: Further collaborator UO4 - RTD-B in Electronic and Computer Bioengineering

Country of residence: ITALY

Nationality: italiana

Country of Birth: RUSSIAN FEDERATION

Date of birth: 11/01/1978

Place of Birth: Tambov

Official H index (Scopus or Web of Science): 14.0

Scopus Author Id:6701679186

ORCID ID:0000-0002-9031-0672

RESEARCH ID:AAY-7265-2020

Contact address

Current organisation name: DIMES Università della Calabria

Current Department / Faculty / Institute / Laboratory name: Department of Electronic and Computer Bioengineering

Street: via P.Bucci, cubo 42C

Postcode / Cedex: 87027

Town: Rende

Phone:+393666461211

Phone 2:

Education / training				
Educational institution and location	Degree	Field of study	From year	To year
Università della Calabria	PhD	Systems Engineering and Computer Science	2001	2005
Università della Calabria	Bachelor Degree / Laurea Triennale	Computer Engineering	1999	2001

Personal Statement:

Irina Trubitsyna is a RTD-B at Università della Calabria. Her role will be to develop an AI model to predict lung cancer risk based on clinical and imaging data in collaboration with UO3.

Positions and honors

Positions					
Institution	Division / Research group	Location	Position	From year	To year
Università della Calabria	Dipartimento di Ingegneria Informatica, Modellistica, Elettronica e Sistemistica	Rende (Cosenza)	RTD-B	2020	0
Università della Calabria	Dipartimento di Ingegneria Informatica, Modellistica, Elettronica e Sistemistica	Rende (Cosenza)	Researcher PTA	2009	2020
Università della Calabria	Dipartimento di Ingegneria Informatica, Modellistica, Elettronica e Sistemistica	Rende (Cosenza)	Researcher and lecturer	2005	2009

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Applicant Institution: Ospedale San Raffaele - Milano	Applicant/PI Coordinator: CHITI ARTURO

Other awards and honors

NA

Grant						
Funded by Institution	Researcher inst. where grant is/was performed	Year	Title	Position in Projects	Fund (euro)	Source website grant listed
Ministero dello Sviluppo Economico	Università della Calabria	2021 - 2022	Progetto TRUE DETECTIVE 4.0	Coordinator	120.000,00	https://magazine.relatech.com/progettotruedetective

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Applicant Institution: Ospedale San Raffaele - Milano	Applicant/PI Coordinator: CHITI ARTURO

2.7 Research Collaborators n. 5

Last Name: Guarrasi

First Name: Valerio

Last name at birth:

Gender: M

Title: Further collaborator UO3 - Postdoc at Università Campus Bio-Medico di Roma

Country of residence: ITALY

Nationality: Italia

Country of Birth: ITALY

Date of birth: 21/04/1995

Place of Birth: Palermo

Official H index (Scopus or Web of Science): 6.0

Scopus Author Id:57219750978

ORCID ID:0000-0002-1860-7447

RESEARCH ID:GWU-9478-2022

Contact address

Current organisation name: Campus Biomedico - Roma

Current Department / Faculty / Institute / Laboratory name: Department of Engineering

Street: Via Iberia 47

Postcode / Cedex: 00183

Town: Roma

Phone:+393466095623

Phone 2:

Education / training				
Educational institution and location	Degree	Field of study	From year	To year
La Sapienza Università di Roma	PhD	Data Science	2019	2022
La Sapienza Università di Roma	Master's Degree / Laurea Magistrale	Data Science	2017	2019
La Sapienza Università di Roma	Bachelor Degree / Laurea Triennale	Management Engineering	2014	2017

Personal Statement:

Valerio Guarrasi is a post-doctoral research fellow at Università Campus Bio-Medico di Roma. His role will be to develop an AI model to predict lung cancer risk based on clinical and imaging data in collaboration with UO4.

Positions and honors

Positions					
Institution	Division / Research group	Location	Position	From year	To year
Università Campus Bio-Medico di Roma	Department of Engineering	Roma	PostDoc	2023	0

Other awards and honors

Winner of the international competition: "Covid CXR Hackaton", Bracco Imaging, Dubai Expo 2020 (2022); winner of the international competition: "AI against COVID-19: Screening X-ray Images for COVID-19 Infection", IEEE SIGHT, University of Waterloo (2021); winner of the Honors Program for the Master's Degree in Data Science (2019); winner of the

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international competition: ζ Stats Under the Stars 4 Competition ζ , Società Italiana Statistica, Findomestic (2018).

Grant						
Funded by Institution	Researcher inst. where grant is/was performed	Year	Title	Position in Projects	Fund (euro)	Source website grant listed
Ministero dello Sviluppo Economico (MISE)	Bracco Imaging	2022	Piattaforma per la Medicina di Precisione. Intelligenza Artificiale e Diagnostica Clinica Integrata	Collaborator	150.000,00	NA
Fondation C.M. Lerici	Umeå University	2022	Multi-expert system for the differentiation between subjects affected by Higher-Level Gait Disorder	Coordinator	2.500,00	https://cmleerici.wordpress.com/borse-di-studio/
La Sapienza Università di Roma	La Sapienza Università di Roma	2022	A network-based approach for in-silico identification of drug-repurposing opportunities for the treatment of neurological diseases	Collaborator	3.167,00	https://www.uniroma1.it/it/pagina/bandi-sapienza-la-ricerca
La Sapienza Università di Roma	La Sapienza Università di Roma	2021	Optimization of Neural Networks for the Diagnosis of COVID-19 on Chest X-ray	Coordinator	1.000,00	https://www.uniroma1.it/it/pagina/bandi-sapienza-la-ricerca
La Sapienza Università di Roma	La Sapienza Università di Roma	2021	Development of a network theory-based algorithm for drug repositioning and its application to COVID-19	Collaborator	12.000,00	https://www.uniroma1.it/it/pagina/bandi-sapienza-la-ricerca
La Sapienza Università di Roma	La Sapienza Università di Roma	2020	Radiomics and Deep Learning for the Detection of COVID-19	Coordinator	1.000,00	https://www.uniroma1.it/it/pagina/bandi-sapienza-la-ricerca
La Sapienza Università di Roma	La Sapienza Università di Roma	2019	International Thesis Scholarship: Generalized Model for Temporal Motifs	Coordinator	3.000,00	https://www.uniroma1.it/en/pagina/international-thesis-scholarship

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2.8 Research Collaborators n. 6 - Under 40

Last Name: Cuocolo

First Name: Renato

Last name at birth:

Gender: M

Title: Responsible UO2 - Associate Professor at Università degli studi di Salerno

Country of residence: ITALY

Nationality: Italiana

Country of Birth: ITALY

Date of birth: 21/04/1986

Place of Birth: Napoli

Official H index (Scopus or Web of Science): 22.0

Scopus Author Id:55253274100

ORCID ID:0000-0002-1452-1574

RESEARCH ID:G-3147-2018

Contact address

Current organisation name: Ospedale San Giovanni di Dio e Ruggi d'Aragona

Current Department / Faculty / Institute / Laboratory name: Diagnostic imaging department

Street: Largo Città Ippocrate

Postcode / Cedex: 84131

Town: Salerno

Phone:+393393444054

Phone 2:

Education / training				
Educational institution and location	Degree	Field of study	From year	To year
Università degli studi di Napoli Federico II	PhD	Scienze Biomorfologiche e Chirurgiche	2017	2019
Università degli studi di Napoli Federico II	Specialization / Specializzazione	Diagnostic Imaging	2012	2017
Università degli studi della Campania ̀Luigi Vanvitellì	Single-cycle master's degree / Laurea magistrale a ciclo unico	Medicine and Surgery	2004	2011

Personal Statement:

Renato Cuocolo is a radiologist at Azienda Ospedaliera Universitaria San Giovanni di Dio e Ruggi d'Aragona and Associate Professor at Università di Salerno. His research interests include the application of machine learning techniques to medical imaging and advanced image analysis in various oncological fields. His role will be to coordinate the activities of UO2 and to collaborate with the other UOs involved in the project.

Positions and honors

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Positions					
Institution	Division / Research group	Location	Position	From year	To year
Università degli studi di Salerno	Department of Medicine, Surgery, and Dentistry Scuola Medica Salernitana	Baronissi (SA)	Associate Professor	2022	0
Università degli studi di Napoli Federico II	Department of Advanced Biomedical Sciences	Napoli	Research Fellow (RTDa)	2020	2022

Other awards and honors

Award for "T2 Mapping of the Sacroiliac Joints at 1.5 T can Identify Patients with Spondyloarthritis" and "ML Classification of Spinal Lesions: Compared Accuracy of Texture Parameters Extracted by Different Software" at MSK SIRM (2019); award for "Classificazione dei condrosarcomi ossei mediante parametri di radiomica estratti con algoritmi di ML" at SIRM (2019); award for "Prediction of high proliferative index in pituitary macroadenomas using MRI-based radiomics and ML" at ESNR (2022)

Grant						
Funded by Institution	Researcher inst. where grant is/was performed	Year	Title	Position in Projects	Fund (euro)	Source website grant listed
European Union's Horizon Europe programme	Università degli Studi di Salerno	2023	COMputational Models FOR patient stratification in urologic cancers: Creating robust and trustworthy multimodal AI for health care (COMFORT)	Collaborator	75.333,75	https://www.comfort-ai.eu/

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Applicant Institution: Ospedale San Raffaele - Milano	Applicant/PI Coordinator: CHITI ARTURO

2.9 Additional Research Collaborators n. 2 - Under 40 to hire

Last Name: DE GIORGI	Last name at birth:
First Name: MARCO	Gender: M
Title: Radiologist	Country of residence: ITALY
Nationality: Italiana	Country of Birth: ITALY
Date of birth: 12/09/1992	Place of Birth: Salerno
Official H index (Scopus or Web of Science): 3.0	
Scopus Author Id: 57196235576	ORCID ID: 0009-0005-0797-0100 RESEARCH ID: NA

Contact address

Current organisation name: Ospedale San Giovanni di Dio e Ruggi d'Aragona
Current Department / Faculty / Institute / Laboratory name: Diagnostic imaging department
Street: Umberto I Nocera inferiore (ASL SALERNO)
Postcode / Cedex: 84131
Phone: +393932094879
Town: Salerno
Phone 2:

Education / training				
Educational institution and location	Degree	Field of study	From year	To year
Università degli Studi di Napoli Federico II	Specialization / Specializzazione	Diagnostic Imaging	2019	2023
Università degli studi di Napoli Federico II	Single-cycle master's degree / Laurea magistrale a ciclo unico	Medicine and Surgery	2010	2016

Personal Statement:
Marco De Giorgi is a radiologist and his role will be to evaluate and interpret LDCT screening images of patients enrolled in UO2.

Positions and honors

Positions					
Institution	Division / Research group	Location	Position	From year	To year
Policlinico Ospedaliero Umberto I di Nocera Inferiore	Department of Diagnostic Imaging	Nocera Inferiore (Salerno)	Radiology consultant	2023	0

Other awards and honors
na

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Applicant Institution: Ospedale San Raffaele - Milano	Applicant/PI Coordinator: CHITI ARTURO

Grant						
Funded by Institution	Researcher inst. where grant is/was performed	Year	Title	Position in Projects	Fund (euro)	Source website grant listed
NA	NA	NA	NA	Coordinator	0,00	NA

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Applicant Institution: Ospedale San Raffaele - Milano	Applicant/PI Coordinator: CHITI ARTURO

2.10 Additional Research Collaborators n. 3 - Under 40 to hire

Last Name: PONSIGLIONE First Name: ANDREA	Last name at birth: Ponsiglione Gender: M
Title: Radiologist Nationality: Italiana Date of birth: 23/03/1988	Country of residence: ITALY Country of Birth: ITALY Place of Birth: Napoli
Official H index (Scopus or Web of Science): 15.0	
Scopus Author Id: 56964476900	ORCID ID: 0000-0002-0105-935X RESEARCH ID: AAR-3677-2020
Contact address	
Current organisation name: Ospedale San Giovanni di Dio e Ruggi d'Aragona	
Current Department / Faculty / Institute / Laboratory name: Diagnostic imaging department	
Street: Via Pansini 5	
Postcode / Cedex: 80131	Town: Napoli
Phone: +393280876517	Phone 2:

Education / training				
Educational institution and location	Degree	Field of study	From year	To year
Università degli Studi di Napoli Federico II	PhD	Biomorphological and Surgical Sciences	2018	2022
Università degli Studi di Napoli Federico II	Specialization / Specializzazione	Diagnostic Imaging	2013	2018
Università degli Studi di Napoli Federico II	Single-cycle master's degree / Laurea magistrale a ciclo unico	Medicine and Surgery	2006	2012

Personal Statement:

Andrea Ponsiglione is a radiologist and post-doctoral research fellow. His role will be to evaluate and interpret LDCT screening images of patients enrolled in UO2.

Positions and honors

Positions					
Institution	Division / Research group	Location	Position	From year	To year
Paul Strickland Scanner Center-Mount Vernon Cancer Center	Diagnostic Imaging	Northwood, United Kingdom	PostDoc exchange	2023	2023
Università degli Studi di Napoli Federico II	Diagnostic Imaging	Napoli	PostDoc	2022	0

Other awards and honors

Sent date: 11/07/2023 13.03

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Bracco Fellowship (2023); invest in the Youth Award, European Congress of Radiology (2020); travel student Award, Annual Meeting of Radiological Society of North America (2018).

Grant						
Funded by Institution	Researcher inst. where grant is/was performed	Year	Title	Position in Projects	Fund (euro)	Source website grant listed
NA	NA	NA	NA	Coordinator	0,00	NA

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Applicant Institution: Ospedale San Raffaele - Milano	Applicant/PI Coordinator: CHITI ARTURO

2.11 Additional Research Collaborators n. 4 - Under 40 to hire

Last Name: PERILLO	Last name at birth:
First Name: TERESA	Gender: F
Title: Radiologist	Country of residence: ITALY
Nationality: Italiana	Country of Birth: ITALY
Date of birth: 31/01/1992	Place of Birth: Villaricca
Official H index (Scopus or Web of Science): 8.0	
Scopus Author Id: 55655979800	ORCID ID: 0000-0002-3840-7637 RESEARCH ID: NA
Contact address	
Current organisation name: Ospedale San Giovanni di Dio e Ruggi d'Aragona	
Current Department / Faculty / Institute / Laboratory name: Diagnostic imaging department	
Street: Via Alfonso de Nicola	
Postcode / Cedex: 84014	Town: Nocera Inferiore
Phone: +393317037575	Phone 2:

Education / training				
Educational institution and location	Degree	Field of study	From year	To year
Università degli studi di Napoli Federico II	Specialization / Specializzazione	Diagnostic Imaging	2017	2021
Università degli studi di Napoli Federico II	Single-cycle master's degree / Laurea magistrale a ciclo unico	Medicine and surgery	2010	2016

Personal Statement:

Teresa Perillo is a radiologist and her role will be to evaluate and interpret LDCT screening images of patients enrolled in UO2.

Positions and honors

Positions					
Institution	Division / Research group	Location	Position	From year	To year
Presidio Ospedaliero Umberto I	Neuroradiology Unit	Nocera Inferiore (SA)	Consultant in neuroradiology (Medical Director Level I)	2022	0

Other awards and honors

NA

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Grant						
Funded by Institution	Researcher inst. where grant is/was performed	Year	Title	Position in Projects	Fund (euro)	Source website grant listed
NA	NA	NA	0	Coordinator	0,00	NA

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2.17 Expertise Research Collaborators

Selected peer-reviewed publications of the Research Group / Collaborators									
Collaborato	Title	Type	Pag	Vol	Year	DOI	PMID	Cit.**	P.*
Trubitsyna Irina	Checking chase termination: Cyclicity analysis and rewriting techniques	Article	621-635	27	2015	10.1109/TKDE.2014.2339816	NOT_FOUND	17	L
Guarrasi Valerio	Pareto optimization of deep networks for COVID-19 diagnosis from chest X-rays	Article	NOT_FOUND	121	2022	10.1016/j.patcog.2021.108242	NOT_FOUND	13	F
PONSIGLIONE ANDREA	Clinically significant prostate cancer detection with biparametric MRI: A systematic review and meta-analysis	Review	608-621	216	2021	10.2214/AJR.20.23219	33502226	15	C
PONSIGLIONE ANDREA	Clinically significant prostate cancer detection on MRI: A radiomic shape features study	Article	144-149	116	2019	10.1016/j.ejrad.2019.05.006	31153556	44	O
PONSIGLIONE ANDREA	Spectrum of lytic lesions of the skull: a pictorial essay	Review	845-856	9	2018	10.1007/s13244-018-0653-y	NOT_FOUND	22	O
VELTRI PIERANGELO	Kit ^{cre} knock-in mice fail to fate-map cardiac stem cells	Article	E1-E17	555	2018	10.1038/nature25771	29565363	36	O
VELTRI PIERANGELO	Adult cardiac stem cells are multipotent and robustly myogenic: C-kit expression is necessary but not sufficient for their identification	Article	2101-2116	24	2017	10.1038/cdd.2017.130	28800128	85	O
VELTRI PIERANGELO	A Multi-Task Framework for Monitoring Health Conditions via Attention-based Recurrent Neural Networks	Article	1665-1674	2017	2017	NOT_FOUND	29854237	29	O
VELTRI PIERANGELO	Intelligent healthcare informatics in big data era	editorial	75-77	65	2015	10.1016/j.artmed.2015.08.002	26306669	26	L
VELTRI PIERANGELO	In PCOS patients the addition of low-dose spironolactone induces a more marked reduction of clinical and biochemical hyperandrogenism than metformin alone	Article	132-139	24	2014	10.1016/j.numecd.2013.04.016	23845740	28	O
Soda Paolo	Benchmarking HEp-2 cells classification methods	Article	1878-1889	32	2013	10.1109/TMI.2013.2268163	23797238	202	O
Soda Paolo	AlforCOVID: Predicting the clinical outcomes in patients with COVID-19 applying AI to chest-X-rays. An Italian multicentre study	Article	NOT_FOUND	74	2021	10.1016/j.media.2021.102216	34492574	27	F

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Collaborato	Title	Type	Pag	Vol	Year	DOI	PMID	Cit.**	P.*
Soda Paolo	Twitter rumour detection in the health domain	Article	33-40	110	2018	10.1016/j.eswa.2018.05.019	NOT_FOUND	61	L
Soda Paolo	ECG databases for biometric systems: A systematic review	Review	189-202	67	2017	10.1016/j.eswa.2016.09.030	NOT_FOUND	81	O
Soda Paolo	A survey on using domain and contextual knowledge for human activity recognition in video streams	Review	97-111	63	2016	10.1016/j.eswa.2016.06.011	NOT_FOUND	60	O
Cuocolo Renato	Machine learning for the identification of clinically significant prostate cancer on MRI: a meta-analysis	Article	6877-6887	30	2020	10.1007/s00330-020-07027-w	32607629	35	F
Cuocolo Renato	Machine Learning in oncology: A clinical appraisal	Review	55-62	481	2020	10.1016/j.canlet.2020.03.032	32251707	49	F
Cuocolo Renato	Prediction of high proliferative index in pituitary macroadenomas using MRI-based radiomics and machine learning	Article	1365-1373	61	2019	10.1007/s00234-019-02266-1	31375883	44	C
Cuocolo Renato	Prostate MRI radiomics: A systematic review and radiomic quality score assessment	Review	NOT_FOUND	129	2020	10.1016/j.ejrad.2020.109095	32531722	50	C
Cuocolo Renato	Machine learning applications in prostate cancer magnetic resonance imaging	Review	NOT_FOUND	3	2019	10.1186/s41747-019-0109-2	31392526	84	F
VERONESI GIULIA	High-dimensional single cell analysis identifies stemlike cytotoxic CD8 ⁺ T cells infiltrating human tumors	Article	2520-2535	215	2018	10.1084/JEM.20180684	30154266	181	O
VERONESI GIULIA	Exposure to low dose computed tomography for lung cancer screening and risk of cancer: Secondary analysis of trial data and risk-benefit analysis	Article	NOT_FOUND	356	2017	10.1136/bmj.j347	28179230	155	O
VERONESI GIULIA	MiR-test: A blood test for lung cancer early detection	Article	NOT_FOUND	107	2015	10.1093/jnci/djv063	25794889	192	O
VERONESI GIULIA	2nd ESMO consensus conference on lung cancer: Early-stage non-small-cell lung cancer consensus on diagnosis, treatment and follow-up	Article	1462-1474	25	2014	10.1093/annonc/mdu089	24562446	292	O
VERONESI GIULIA	Robotic lobectomy for non-small cell lung cancer (NSCLC): Long-term oncologic results	Article	383-389	143	2012	10.1016/j.jtcvs.2011.10.055	22104677	190	L

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Collaborato	Title	Type	Pag	Vol	Year	DOI	PMID	Cit.**	P.*
Ferrara Roberto	Hyperprogressive Disease in Patients with Advanced Non-Small Cell Lung Cancer Treated with PD-1/PD-L1 Inhibitors or with Single-Agent Chemotherapy	Article	1543-1552	4	2018	10.1001/jamaoncol.2018.3676	30193240	449	F
Ferrara Roberto	Association of the lung immune prognostic index with immune checkpoint inhibitor outcomes in patients with advanced non-small cell lung cancer	Article	351-357	4	2018	10.1001/jamaoncol.2017.4771	29327044	471	O
Ferrara Roberto	Hyperprogressive disease: recognizing a novel pattern to improve patient management	Review	748-762	15	2018	10.1038/s41571-018-0111-2	30361681	255	O
Ferrara Roberto	Patterns of responses in metastatic NSCLC during PD-1 or PDL-1 inhibitor therapy: Comparison of RECIST 1.1, irRECIST and iRECIST criteria	Article	38-47	88	2018	10.1016/j.ejca.2017.10.017	29182990	210	O
Ferrara Roberto	Impact of baseline steroids on efficacy of programmed cell death-1 and programmed death-ligand 1 blockade in patients with non-small-cell lung cancer	Article	2872-2878	36	2018	10.1200/JCO.2018.79.0006	30125216	569	O

* Position: F=First L=Last C=Correspondent O=Other N=Not applicable

** Autocertificated

3 - Ethics

1. HUMAN EMBRYOS/FOETUSES	
Does your research involve Human Embryonic Stem Cells (hESCs)?	No
Does your research involve the use of human embryos?	No
Does your research involve the use of human foetal tissues / cells?	No
2. HUMANS	
Does your research involve human participants?	Yes
Does your research involve physical interventions on the study participants?	No
3. HUMAN CELLS / TISSUES	
Does your research involve human cells or tissues (other than from Human Embryos/ Foetuses)?	No
4. PERSONAL DATA	
Does your research involve personal data collection and/or processing?	Yes

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Does your research involve further processing of previously collected personal data (secondary use)?	No
5. ANIMALS	
Does your research involve animals?	No
6. ENVIRONMENT & HEALTH and SAFETY	
Does your research involve the use of elements that may cause harm to the environment, to animals or plants?	No
Does your research deal with endangered fauna and/or flora and/or protected areas?	No
Does your research involve the use of elements that may cause harm to humans, including research staff?	No
7. DUAL USE	
Does your research involve dual-use items in the sense of Regulation 428/2009, or other items for which an	No
8. EXCLUSIVE FOCUS ON CIVIL APPLICATIONS	
Could your research raise concerns regarding the exclusive focus on civil applications?	No
9. MISUSE	
Does your research have the potential for misuse of research results?	No
10. OTHER ETHICS ISSUES	
Are there any other ethics issues that should be taken into consideration? Please specify	No

I confirm that I have taken into account all ethics issues described above and that, if any ethics issues apply, I will complete the ethics self-assessment and attach the required documents.

☒

4 - Call-specific questions

Eligibility	
I acknowledge that I am aware of the eligibility requirements for applying as specified in the Call-PNRRXXXX_M6/C2, and certify that, to the best of my knowledge my application is in compliance with all these requirements. I understand that my proposal may be declared ineligible at any point during the evaluation or granting process if it is found not to be compliant with these eligibility criteria.	☒
I confirm that the proposal that I am about to submit draws substantially don't repeat on an existing or recently finished GRANT funded.	☒

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Data-Related Questions and Data Protection (Consent to any question below is entirely voluntary. A positive or negative answer will not affect the evaluation of your project proposal in any form and will not be communicated to the evaluators of your project.)	
For communication purposes only, the MoH asks for your permission to publish, in whatever form and medium, your name, the proposal title, the proposal acronym, the panel, and host institution, should your proposal be retained for funding.	☒
Some national and regional public research funding authorities run schemes to fund MoH applicants that score highly in the MoH's evaluation but which can not be funded by the MoH due to its limited budget. In case your proposal could not be selected for funding by the MoH do you consent to allow the MoH to disclose the results of your evaluation (score and ranking range) together with your name, non-confidential proposal title and abstract, proposal acronym, host institution and your contact details to such authorities?	☒
The MoH is sometimes contacted for lists of MoH funded researchers by institutions that are awarding prizes to excellent researchers. Do you consent to allow the MoH to disclose your name, non-confidential proposal title and abstract, proposal acronym, host institution and your contact details to such institutions?	☒
The Ministry of Health occasionally could contact Principal Investigators of funded proposals for various purposes such as communication campaigns, pitching events, presentation of their project's evolution or outcomes to the public, invitations to represent the Ministry of Health in national and international forums, studies etc. Should your proposal be funded, do you consent to the Ministry of Health staff contacting you for such purposes?	☒
For purposes related to monitoring, study and evaluating implementation of MoH actions, the MoH may need that submitted proposals and their respective evaluation data be processed by external parties. Any processing will be conducted in compliance with the requirements of Regulation 45/2001.	

5 – Description Project

Summary description

Currently available screening programmes for lung cancer are limited by many challenges including low diagnostic accuracy, radiation exposure and high costs. New technologies in PET/CT scanners can allow cheaper and more sensitive exams with low radiation exposure. AI can be used to denoise LDCT to enhance the accuracy of imaging tests and build risk-assessment models. This project aims to develop a new approach exploiting both these revolutionary advancements to bridge the existing gap in lung cancer screening. Patients in a high-risk population will be enrolled into two different cohorts undergoing LDCT scan and simultaneous [18F]FDG PET/CT on new-generation long axial field of view scanner (UO1) or screening with low LDCT only (UO2). AI will assist in image enhancement and interpretation and will develop a personalised risk-model guiding the following steps of clinical management, significantly improving early diagnosis of lung cancer, reducing mortality and healthcare costs.

Background / State of the art and Preliminary data (if available)

Lung cancer is the leading cause of cancer-related morbidity and mortality worldwide. Early diagnosis has the greatest impact on clinical outcome, with a 5-year survival rate of over 75% (Goldstraw et al, 2016). Along with smoking cessation, early detection through screening of the asymptomatic high-risk population with low dose computed tomography (LDCT) is now the main strategy to significantly reduce mortality by up to 50%. Widespread implementation of lung cancer screening faces several challenges: high inter- and intra-observer variability, risk of overdiagnosis and false-positive findings, increase in unnecessary invasive procedures, lack of consensus on optimal timing of imaging intervals, radiation exposure and economic burden on the healthcare system (Novellis et al, 2016). Individual risk prediction models can improve the selection

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of patients for screening (Tammemägi et al, 2013). Positron Emission Tomography (PET) with 2-[fluorine-18]-fluoro-2-deoxy-d-glucose ([18F]FDG) is a non-invasive imaging modality that plays a key role in the management of patients with oncological malignancies including lung cancer. [18F]FDG PET/CT is highly accurate in the differential diagnosis of indeterminate lung nodules on CT (Wi et al, 2015). New technologies in PET scanners and reconstruction algorithms allow a reduction in dose and acquisition time, ensuring excellent image quality with limited radiation exposure and costs (Aide et al, 2021).

Description and distribution of activities of each operating unit

IRCCS Ospedale San Raffaele (UO1) is an internationally renowned, highly specialised university hospital that provides advanced care and contributes to the development of new therapies for a wide range of diseases. UO1 offers a multidisciplinary approach and the latest technology on an everyday basis to provide earlier diagnosis and a more personalised approach to treatment. UO1 will be the leader and coordinator of the planned activities of the study between the different UOs involved. The research team of UO1 will consist of nuclear medicine physicians, thoracic surgeons and oncologists who will work together in multidisciplinary meetings to select and manage the patients who will be enrolled in the study. The Department of Thoracic Surgery of UO1 will be responsible for recruiting patients eligible for low dose [18F]FDG PET/CT screening. The Department of Nuclear Medicine of UO1 will be responsible for managing the flow of patients eligible for screening, performing the low dose [18F]FDG PET/CT scans and evaluating and interpreting the [18F]FDG PET/CT images. Azienda Ospedaliera Universitaria San Giovanni di Dio e Ruggi d'Aragona (UO2) is a hospital closely linked to the Università di Salerno, in particular the Faculties of Medicine, Surgery and Dentistry. The research team of UO2 will consist of highly specialised and experienced radiologists who will work with clinicians to identify patients to be recruited for LDCT screening. The Department of Diagnostic Imaging of UO2 will be responsible for enrolling patients, managing patient flow for the examination, performing the LDCT examination, and evaluating and interpreting the LDCT images. Università Campus Bio-Medico di Roma (UO3) is an internationally renowned institution: its research team will be composed of biomedical engineers with a high scientific profile and specific technical-scientific and engineering skills in AI, deep learning and medical imaging, as well as with transversal skills of medical-biological interest acquired through daily close interdisciplinary collaboration with medical departments. Università della Calabria (UO4) is a centre of academic excellence with a strong multidisciplinary commitment. The UO4 research team will consist of electronic and computational bioengineers with extensive experience in translational projects in the healthcare field. Biomedical engineers from UO3 and UO4 will work with clinicians of UO1 and UO2 to develop an AI model to predict the risk of lung cancer based on clinical and imaging data and to deploy tailored screening programmes according to risk stratification. In particular, UO3 is committed with image denoising and virtual scanning, and it cooperates with UO4 in multimodal learning, exploiting also explainable AI (XAI) to highlight fairly and transparently which are morphological and metabolic features predicting lung cancer risk. With reference to denoising, we will exploit previous results from UO3 that uses Cycle Generative Adversarial Network (CycleGAN), which is a technique that involves the automatic training of image-to-image translation models without paired examples. As for virtual scanning, a similar technique will be employed, comparing also with pix2pix, a conditional GAN that learns a mapping from input to output images. For multimodal learning we will use joint, early, late methods working in conjunction with post-hoc XAI approaches, both developed within UO3 and UO4.

5.4 Specific Aims and Experimental Design

Specific aim 1

Assessment of the potential added value of low dose [18F]FDG PET/CT in the early detection of lung cancer in the screening work-up of the high-risk population. The target population (cohort 1) for the multimodal screening programme will be identified at UO1 on the basis of the PLCOM2012 prediction risk model. Selection criteria will include age, ethnicity, education, body mass index, history of chronic obstructive pulmonary disease, history of cancer, family history of lung cancer, smoking status and behaviour, and years of cessation for former smokers. Exclusion criteria will be ongoing pregnancy and breastfeeding according to standard clinical practice, previous diagnosis of lung cancer, previous CT scan within the last 24 months, and concomitant severe clinical conditions. All patients enrolled in cohort 1 will undergo a LDCT

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and a simultaneous low dose [18F]FDG PET scan. A latest-generation long axial field of view PET/CT scanner will be used for whole-body image acquisition. This cutting-edge technology will reduce the injected [18F]FDG activity and therefore the radiation exposure, while providing high quality images. Experienced nuclear medicine physicians will evaluate and interpret [18F]FDG PET/CT images. Patients will be managed according to clinical practice (i.e., evolutive monitoring, histological confirmation with invasive procedures and/or surgery to remove the lesion) if suspicious findings are identified on [18F]FDG PET/CT images.

Specific aim 2

Assessment of the role of LDCT alone in the early detection of lung cancer in the screening work-up of the high-risk population. The control population (cohort 2) will be identified in UO2 on the basis of the PLCOM2012 prediction risk model as previously described for cohort 1. Exclusion criteria will be ongoing pregnancy, previous diagnosis of lung cancer, previous CT scan within the last 24 months, and concomitant severe clinical conditions. All patients enrolled in cohort 2 will undergo LDCT examination. Experienced radiologists will evaluate and interpret LDCT images. Patients will be managed according to clinical practice (i.e., evolutive monitoring, histological confirmation with invasive procedures and/or surgery to remove the lesion) if suspicious findings are identified on LDCT.

Specific aim 3

Development of an Artificial Intelligence (AI) model based on comprehensive data to predict lung cancer risk in the high-risk lung cancer screening population. The AI model will combine patient-specific data, including individual risk factors and baseline clinical characteristics, with imaging features derived from LDCT scans and [18F]FDG PET/CT. Different AI models will be developed, trained and tested to perform different tasks: characterisation and classification of lung nodules on LDCT and PET/CT images (1); selection of patients for invasive diagnostic procedures (2); discrimination of patients who can withdraw from the screening programme early and those who need to continue monitoring (3); identification of the most appropriate imaging modality for further non-invasive assessment (i.e., LDCT or PET/CT) via the use of generative approaches (4); individualised scheduling of screening intervals and duration (5); identification of the target population for the screening programme based on individual risk factors and baseline clinical characteristics (6).

Experimental design aim 1

Legal, privacy and ethical requirements will be addressed before the project starts, under the responsibility of UO1 (WP1). We will prospectively enrol a primary cohort of patients at high risk of lung cancer (cohort 1) at UO1, who will undergo low-dose PET/CT for screening (WP2.1; WP2.2). The target population will be identified based on lung cancer risk prediction using the PLCOm2012 model, including age, ethnicity, education, body mass index, history of chronic obstructive pulmonary disease, history of cancer, family history of lung cancer, smoking status, average number of cigarettes per day, duration of smoking and years of cessation for former smokers. Inclusion criteria will be age > 50 years, PLCOm2012 risk prediction > 4%, and signed written informed consent form. Exclusion criteria will be blood glucose levels <200 mg/dl, ongoing pregnancy and breastfeeding, unwillingness to participate, previous diagnosis of lung cancer, previous CT scan within the last 24 months, concomitant severe clinical conditions and any condition that preclude the feasibility of the study. We plan to include in the study 167 patients in cohort 1. Epidemiological, clinical, and smoking status data included in the PLCO m2012 model will be collected (WP2.2.1).

All patients enrolled in cohort 1 will undergo a LDCT and a simultaneous low dose [18F]FDG PET scan (WP3.1). Whole-body [18F]FDG PET/CT will be performed according to EANM Guidelines for tumour imaging (Boellaard R et al, Eur J Nucl Med 2015). Briefly, patients will be instructed to fast for 6 hours and to avoid strenuous physical activity for 48 hours prior to the PET/CT scan. Diabetic patients will be instructed to fast for 4 hours prior to the scan and will be instructed on the use of medications according to the institution's protocol. Blood glucose levels will be checked, [18F]FDG will be injected intravenously, and PET/CT images will be acquired after 60 minutes ($\pm$ 10 minutes). A new generation long axial field of view PET/CT scanner (Omni Legend, General Electric Healthcare, Waukesha, WI, USA) will be used for whole body imaging. A low dose [18F]FDG PET/CT acquisition protocol will be defined prior to the start of the project.

PET/CT images will be qualitatively assessed independently by at least two experienced nuclear medicine physicians

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(WP3.1). PET/CT images will be first analysed with the lung window to detect any findings suggestive of a lung tumour. Whole-body PET/CT images will be then analysed to detect any incidental finding in the chest as well as other anatomical areas included in the field of view. PET/CT images will be considered positive if there is at least one non-calcified lung nodule or any suspicious finding on CT scan characterised by focal [18F]FDG uptake deviating from physiological distribution or above physiological background activity. If there are no areas of increased uptake compared to physiological background activity, PET/CT images will be considered negative. The number, size, characteristics, and location of any lesion detected will be recorded for each PET/CT scan. Any finding suggestive of clinically significant conditions other than cancer will be reported. Sensitivity, specificity, and accuracy will be calculated by per-patient and per-lesion analysis (WP3.2).

Patients will be managed according to good clinical practice (i.e., 1-/3-months follow-up, CT-guided biopsy for histological confirmation, surgery) if suspicious findings are identified on [18F]FDG PET/CT images. Patient management will be discussed during the multidisciplinary board. Follow-up data will be collected, including clinical and imaging data, biopsy and/or surgical data, histology, and tumour staging according to TNM classification 8th edition (WP2.2.2). Final diagnosis will be established according to histopathology, if available. Alternatively, clinical and/or imaging follow-up will be used.

Experimental design aim 2

We will prospectively enrol a validation cohort of patients at high risk for lung cancer (cohort 2) at UO2, who will undergo LDCT for screening (WP2.1; WP2.2). The same inclusion/exclusion criteria described for the cohort 1 will be used to select eligible patients. We plan to include in the study 167 patients in cohort 2. Epidemiological, clinical, and smoking status data included in the PLCO m2012 model will be collected.

All patients enrolled in cohort 2 will undergo LDCT examination (WP3.1). The LDCT scan will be performed in single deep inspiratory breath hold. No intravenous contrast will be administered. LDCT images will be assessed and interpreted by at least two experienced radiologists (WP 3.1). LDCT will be defined as positive if at least one non-calcified lung nodule >5 mm in any diameter is detected. LDCT will be defined as negative if no clinically significant morphological alterations are detected. Non-calcified nodules of 3-5mm detected by LDCT will be reported. Lung-RADS criteria will be used to classify detected lung nodules. Any other suspicious morphological alteration will be reported. The number, size, characteristics, and location of any lesion detected will be recorded for each LDCT scan. Any abnormality suggestive of clinically significant conditions other than cancer will be also reported. Sensitivity, specificity, and accuracy will be calculated by per-patient and per-lesion analysis (WP3.2).

Patients will be managed according to good clinical practice (i.e., 1-/3-months follow-up, CT-guided biopsy for histological confirmation, surgery) if suspicious findings are identified on [18F]FDG PET/CT images. Patient management will be discussed during the multidisciplinary board. Follow-up data will be collected, including clinical and imaging data, biopsy and/or surgical data, histology, and tumour staging according to TNM classification 8th edition (WP2.2.2). Final diagnosis will be established according to histopathology, if available. Alternatively, clinical and/or imaging follow-up will be used.

Experimental design aim 3

The AI models will be set up by OU 3 and OU 4 in three sequential work packages (WP4.1-3).

Data pre-processing (WP4.1) will ensure the consistency and quality of the dataset and prepare it for subsequent model development and training. It will consist of: standardisation of patient-specific data and imaging modalities to ensure comparability between different samples and cohorts; handling of missing data by determining the most appropriate method for imputation; extraction of relevant morphological, textural and metabolic features from the pre-processed images, capturing important features of lung nodules and other relevant structures.

AI Model Development and Training (WP4.2) will enable accurate prediction and classification, by learning patterns and relationships between patient-specific data, imaging features, and the desired outcomes. For lung nodule characterization and classification (Task 1), state-of-the-art convolutional neural networks will be trained on the LDCT and PET/CT images. For patient selection for invasive diagnostic procedures (Task 2), ad-hoc multimodal models will be developed and trained using patient-specific data and imaging features, applying suitable methodologies for the considered modalities (e.g., early

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fusion, joint fusion, and late fusion). For patient withdrawal determination (Task 3), as in Task 2, ad-hoc multimodal models will be developed and trained using patient-specific data and imaging features to distinguish between those who can discontinue screening and those who require ongoing monitoring. For the identification of the appropriate imaging modality (Task 4), ad-hoc multimodal models will be developed and trained to assess the suitability of LDCT or PET/CT for further non-invasive assessment based on patient-specific data and imaging features. For individualized screening interval scheduling (Task 5), different machine learning and deep learning techniques (e.g., support vector regressors and multi-layer perceptrons, respectively) will be used to predict the optimal timing and duration of screening based on patient-specific data. For identifying the target population (Task 6), as in Task 5, different machine learning and deep learning techniques will be employed to determine the individuals who are most likely to benefit from the screening program based on their individual risk factors and baseline clinical characteristics. All the models will be incorporated with model-agnostic interpretability techniques (e.g., LIME, SHAP), and visualizations (e.g., heatmaps and bar charts) to provide insights into the decision-making process of the AI models. This will be integrated both in the unimodal and multimodal scenario, to obtain both feature and modality importance, respectively.

Model Evaluation and Comparison (WP4.3) will provide an objective assessment of the predictive performance and identify those with high accuracy and reliability, enabling the selection of the most effective for the intended objectives. To optimize the performance of the models, all the experiments and the hyperparameter tuning will be performed in cross-validation, ensuring rigorous evaluation and a thorough assessment of their generalisability. The generalizability and robustness of the models will be evaluated by testing them on independent test sets defined by k-fold cross validation and on publicly available external data. The performance of different AI models developed for each task will be compared using statistical tests. The most accurate (based on the mean performance) and robust (based on the standard deviation) models will be selected. Considering the trade-offs between model complexity, interpretability, and predictive performance, we can identify the most suited model for its integration into clinical practice. The progress of the project will be constantly monitored. Any difficulties will be discussed within the research team to resolve criticisms and/or identify alternative strategies (WP6).

Picture to support preliminary data

Picture1.jpg

Hypothesis and significance

In the era of personalised medicine, a lung cancer screening programme tailored to each patient's individual risk factors, clinical data and imaging characteristics would perform better than a standardised "one-size-fits-all" strategy. Our central hypothesis is that a comprehensive approach using multimodality imaging with LDCT and low dose [18F]FDG PET/CT integrated with artificial intelligence (AI) models may enable the early detection of suspicious pulmonary nodules in the context of lung cancer screening programme. The combination of morphological and metabolic features in a hybrid PET/CT scanner might improve the diagnostic accuracy for characterising indeterminate lung nodules. Furthermore, whole-body PET/CT images will be able to detect other incidental malignancies with high sensitivity, especially in this selected high-risk patient population. The AI model would combine patient-specific risk factors and clinical data with LDCT and low dose [18F]FDG PET/CT imaging characteristics to select individuals who can drop out the screening programme or who may benefit from additional assessments. Moreover, the AI model would define a personalised screening programme for each patient with indeterminate nodules, directing them to the most appropriate method, i.e., serial LDCT scans, predicting the optimal timing and duration, or referring them directly to PET/CT or invasive procedures.

5.5 Methodologies and statistical analyses

Methods of data collection

The clinical trial will be conducted in accordance with European Union legislation on clinical trials (EU Regulation No. 536/2014), the principles of the ICH Harmonised Tripartite Guidelines for Good Clinical Practice and the ethical principles of the Declaration of Helsinki. All data generated in this research project will be collected in accordance with the European Union General Data Protection Regulation (GDPR) (EU Regulation No. 2016/679. Patients' enrolment will start after the

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Independent Ethics Committee review and approval. Epidemiological, clinical, and imaging data of enrolled patients will be anonymised. Aggregated data will be used for scientific publication and dissemination. A certified electronic case report form (eCRF) will be used to collect data and be shared between participating centres. The raw and reconstructed image data derived from the PET/CT scans will be stored in the institutional Picture Archiving and Communication System (PACS) of UO1. The raw and reconstructed image data derived from the LDCT scans will be stored in the institutional Picture Archiving and Communication System (PACS) of UO2. Clinical and imaging data will be shared anonymously with other centres responsible for creating AI algorithms (UO3 and UO4) through a sharing infrastructure that meets privacy, security, and ethical standards. UO1 will be committed to complying with all legal, privacy and ethical requirements.

Statistic plan

The sample size was determined with reference to the published literature on the specificity of LDCT in lung cancer screening and PET/CT in the differentiation of nodules detected by LDCT in lung cancer screening. Guo L. et al. showed in a meta-analysis that LDCT at baseline had a specificity of 86% for the identification of lung nodules diagnosed as lung cancer using a positivity threshold of 5 mm (Guo L et al, 2023). Veronesi G. et al. showed that PET/CT has a 95% specificity for differentiating benign from malignant lung nodules <10 mm in patients undergoing PET/CT for suspicious findings on lung cancer screening with LDCT (Veronesi G et al, 2015).

The sample size analysis was performed assuming a 9% increase in specificity of PET/CT over LDCT in lung cancer screening. A 95% confidence interval was set with one-sided testing at the 5% level to provide 80% power. The sample size analysis indicated that a total of 303 patients would need to be included to detect a statistically significant difference between LDCT and PET/CT (Akoglu H, 2022). Adjusting the sample size for the prevalence of the disease, which can be estimated to be 5% in selected patients at high risk of developing lung cancer, the total sample size is 318 patients. Taking into account the 5% screening failure, the total sample size is increased to 334. Therefore, the planned enrolment will be 167 patients in each cohort.

Statistical analysis

Baseline patient characteristics including epidemiological, clinical, and smoking status data will be described in frequency tables and descriptive statistics will be provided for quantitative variables. Continuous variables will be expressed as mean and standard deviation if normally distributed and as median and interquartile range if not. Categorical variables will be expressed as frequencies and percentages. [18F]FDG PET/CT and LDCT results will be recorded and cross-tabulated to calculate the diagnostic performance of the two methods. Sensitivity, specificity, and accuracy of [18F]FDG PET/CT and LDCT will be calculated by per-patient and per-lesion separate analyses. The discriminative power of [18F]FDG PET/CT and LDCT will be assessed by calculating the area under the receiver operating characteristic curve (AUC). Fisher's exact test and Chi-squared test will be used to compare differences in categorical variables between groups and to compare the diagnostic performance of [18F]FDG PET/CT and LDCT. Student t-test, Mann-Whitney test and Kruskal-Wallis test will be used to compare differences in continuous variables between groups. Interobserver agreement in lesion detection will be estimated using Cohen's kappa coefficient method. Linear and logistic regression analysis will be used to explore the relationship between clinical/imaging variables and outcome. Machine learning approaches including cluster analysis, principal component analysis, and linear discriminant analysis will be considered to analyse data. Other statistical approaches will be evaluated for small samples or unbalanced groups that may affect power analysis. Appropriate evaluation metrics such as precision, F1 recall, sensitivity, specificity, accuracy, and the AUC will be used to assess the performance of each AI model. Student t-test and Wilcoxon Signed-Rank test will be used to compare the performance of different AI models developed for each task. P-value ≤ 0.05 will be considered for statistical significance.

Timing of analysis data

The proposed project will be a two-year, prospective, multi-centre, open-label clinical trial. Once the project is funded, the necessary documentation will be immediately submitted to the independent Ethics Committee by UO1 for review and approval, which is expected within month 3 (WP1). Prospective enrolment of patients in UO1 and UO2 will begin immediately after ethics committee approval and will run in parallel up to month 18 (WP2.1; WP2.2). Enrolled patients will

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undergo PET/CT or LDCT imaging up to month 18 (WP3.1). Concurrently, baseline clinical data will be collected, and PET/CT and LDCT imaging will be interpreted (WP2.2.1; WP 3.2.1). An image pre-processing method will be developed within month 9 (WP4.1). Artificial Intelligence models will be developed and trained up to month 15 (WP4.2). Artificial Intelligence models will be tested, evaluated, and compared up to month 21 (WP4.3). Follow-up data will be collected up to month 24 (WP2.2.2). Statistical analysis will be performed up to month 24 (WP5.1). Artificial Intelligence models will be validated up within month 24 (WP5.2). Health technology assessment will be performed within month 24 (WP5.3). Interim analyses will be performed at month 6 and 18. The progress of the project will be constantly monitored, and any difficulties will be discussed within the research team to resolve criticisms and/or identify alternative strategies (WP6). Final analysis and final report will be provided within month 24 (WP 7).

5.6 Expected outcomes

We will demonstrate that low dose [18F]FDG PET/CT plays an additional role in screening the high-risk population for lung cancer. [18F]FDG PET/CT will detect lung nodules suspicious for lung cancer with higher specificity compared to LDCT, reducing the number of false positives and overdiagnosis. [18F]FDG PET/CT will provide accurate staging of the disease in terms of both lymph node localisation and metastasis, with significant implications for patient management. Moreover, we will demonstrate that [18F]FDG PET/CT allows the detection of non-lung tumours that cannot be identified by chest LDCT alone, taking advantage of the whole-body scan in a single session. We will develop an Artificial Intelligence model to identify individuals at highest risk who would benefit from lung cancer screening. The Artificial Intelligence model will tailor the screening programme to each individual and guide patient management when imaging shows suspicious findings. We expect this approach to reduce the number of scans, radiation exposure, unnecessary invasive procedures, and therefore healthcare costs.

5.7 Risk analysis, possible problems and solutions

The legal, data protection and ethical requirements will be addressed prior to the start of the project. We do not anticipate any problems in submitting the necessary documentation to the Independent Ethics Committee and obtaining its timely approval.

The screening programme will be promoted to the population through institutional online dissemination channels and through general practitioners in the Omceo registry. We do not anticipate low or delayed patient recruitment for [18F]FDG PET/CT and LDCT for lung cancer screening. IRCCS Ospedale San Raffaele (UO1) is a recognised reference centre for the diagnosis and treatment of lung cancer. A screening programme for lung cancer using LDCT has already been established and is ongoing at the Institute. Azienda Ospedaliera Universitaria San Giovanni di Dio e Ruggi d'Aragona (UO2) has a high patient flow in the diagnostic imaging department. As there are no screening programmes for lung cancer in the area, we expect a high recruitment rate. We expect to obtain ethics committee approval in the first 3 months of the project and to enrol at least 105 patients (approximately 11 patients per month) in each cohort in the first year. Interim analyses will be conducted at 6 and 18 months to assess enrolment rates and to apply any corrective strategies to meet milestone goals. If necessary, sample size will be re-estimated. In the second year, patients' enrolment will continue until month 18. If patient enrolment is insufficient at the 18-month interim analysis, it will be extended up to month 21. This will ensure six months of follow-up data collection and allow for data analysis. The research team of UO1 and UO2 has extensive experience with [18F]FDG PET/CT and CT imaging in oncology, especially in lung cancer. Therefore, we do not expect any difficulty in image evaluation and interpretation.

The amount of clinical data collected, and diagnostic examinations performed in UO1 and UO2 may be inadequate to develop AI models: pre-existing lung cancer screening databases at the Institute and publicly available external data, pre-trained models and/or transfer learning, embeddings and other approaches will be considered to improve the performance of AI models. The research teams at the University Campus Bio-Medico of Rome (UO3) and the University of Calabria (UO4) have extensive experience in the application of Artificial Intelligence and deep learning techniques in medical imaging. Therefore, we don't anticipate any difficulties in data processing and Artificial Intelligence model development. If

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the availability or the quality of training data impacts the performance of an AI model, we will use augmentation techniques to increase the data quantity and quality. For image-based tasks, augmentation techniques can include random rotations, translations, flips (horizontal or vertical), scaling, cropping, brightness/contrast adjustments, and adding noise or blur. These transformations create new training samples that exhibit different variations of the original images, making the model more resilient to different image conditions.

5.8 Significance and Innovation

PET/CT plays a key role in the characterisation of lung nodules and the staging and restaging of lung cancer. This project would pioneer the use of PET/CT in lung cancer screening. Low dose [18F]FDG PET/CT, combining morphological and metabolic imaging, will optimise lung cancer detection and staging while minimising radiation exposure and unnecessary invasive procedures. Smoking also significantly increases the risk of head and neck, gastrointestinal and bladder cancer. PET/CT will detect non-lung cancers that would be missed by chest LDCT alone, taking advantage of the whole-body scan in a single session. AI is emerging as a powerful tool for medical image analysis. We will develop AI models that combine patient-specific data, morphological and metabolic features to predict lung cancer risk and tailor screening programmes. This comprehensive multimodal approach has the potential to increase the effectiveness of lung cancer screening and improve patient outcomes.

5.9 Bibliography

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5.10 Timeline / Deliverables / Payable Milestones

- Milestone 01 (month 3) = Independent Ethics Committee approval (WP 1)

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- Milestone 02 (month 6) = interim analysis (WP 7)
- Milestone 03 (month 9) = image preprocessing method development (WP 4.1)
- Milestone 04 (month 12) = baseline clinical data collection and imaging for all enrolled patients (reasonable enrolment based on activity in each UO is 105 patients) (WP 2.1; WP 2.2; WP 2.2.1; WP 3.1)
- Milestone 05 (month 15) = Artificial Intelligence models development and training (WP 4.2).
- Milestone 06 (month 18) = interim analysis (WP 7); baseline clinical data collection and imaging for all enrolled patients (167 in each UO) (WP 2.1; WP 2.2; WP 2.2.1; WP 3.1)
- Milestone 07 (month 21) = AI models testing, evaluation and comparison (WP 4.3)
- Milestone 08 (month 24) = follow-up data collection (WP 2.2.2); statistical data analysis (WP 5.1); AI models validation (WP 5.2); Health technology assessment (WP 5.2); final analysis and report (WP 7).

Milestones 12 month

WP1: administrative procedures will be completed (within the first 3 months). WP2: clinical data will be collected for all enrolled patients. WP3: image acquisition and interpretation will be completed for all enrolled patients. WP4: image preprocessing method will be developed.

Milestones 24 month

WP2: all patients will be enrolled; clinical data will be collected for all patients. WP3: image acquisition and interpretation will be completed for all enrolled patients. Per lesion and per patient analysis will be completed. WP4: AI models will be developed and trained. AI models will be tested, evaluated, and compared. WP5: statistical analysis, AI model validation and health technology assessment will be completed. WP 6: Risk analysis, quality assurance and data monitoring will be completed. WP 7: Final analysis and final report will be provided.

Gantt chart

GANTT chart.jpg

5.11 Equipment and resources available

Facilities Available

The Nuclear Medicine Unit of UO1 is equipped with a dedicated medical cyclotron, a laboratory for radiopharmaceutical synthesis and quality control, two PET/CT scanners and a new generation long axial field of view PET/CT scanner (Omni Legend, General Electric Healthcare, Waukesha, WI, USA), all accredited with the EARL quality programme of EANM. The institutional RIS-PACS system stores the PET/CT imaging data. The Thoracic Surgery Unit is equipped with the latest instruments for open surgery, minimally invasive video-thoroscopic and robotic surgery, and operative bronchoscopy. The Medical Oncology Unit is equipped for complex treatment pathways using chemo-, immune-, and innovative therapies. UO1 is a reference centre for the diagnosis and treatment of lung cancer. The different departments involved in the management of the disease work closely together and hold regular multidisciplinary meetings. All the facilities of participating departments will be available for the proposed project.

The Diagnostic Imaging Unit of UO2 is equipped with a 256-slice CT scanner (Revolution, General Electric Healthcare, Waukesha, WI, USA).

The Laboratory of Computing Systems and Bioinformatics of UO3 has computational resources for the development of machine learning and deep learning algorithms and applications. The laboratory is acquiring (order placed) a workstation for research in AI with performance in the PetaFLOPS range thanks to 4 NVIDIA A100 Tensor Core 80GB GPUs and 4 NVIDIA A40 48GB GPUs. It is also supplied with an Intel Xeon E5-2697 v2 2.70Ghz 12 Core workstation, 192GB RAM, NVIDIA TESLA K20c 5GB and NVIDIA Quadro K5000 4G, and an INTEL XEON 5450 3.00 12M/1333 Q workstation, 32GB RAM, Nvidia GeForce GTX TITAN 6GB.

The Artificial Intelligence and Data Science Laboratory of UO4 for the development of knowledge and technologies in the fields of Artificial Intelligence, Database and Knowledge-Based Systems, Big Data Management and Analytics, Data Mining

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and Machine/Deep Learning, Information Retrieval, Knowledge Management. The Lab is furnished with 2 servers with 2 Intel Xeon Gold 5118 processors, each with 8 GPU Nvidia 11GB, 8 x 32GB RAM and 4 SSD Intel DC 1TB. One storage unit with 4 SSD/200GB and 20 disks SAS/900 GB.

Subcontract

Due to the experimental design, a Contract Research Organisation (CRO) will be required.

5.12 Desc. of the complementarity and synergy of secondary collab. researchers

Arturo Chiti (UO1) is the director of the Department of Nuclear Medicine at IRCCS Ospedale San Raffaele and Full Professor of Diagnostic Imaging and Radiotherapy at Università Vita-Salute San Raffaele in Milan. He has extensive experience in conducting translational research projects using artificial intelligence techniques in oncology, particularly in lung cancer. His role will be to coordinate the study and supervise all research activities in UO1 and other UOs involved in the project. In addition, he will interpret PET/CT images. Giulia Veronesi (UO1) is the director of the Robotic Thoracic Surgery at the IRCCS Ospedale San Raffaele and Associate Professor at Università Vita-Salute San Raffaele of Milan. She is leading several research projects aimed at developing effective screening programmes to enable early diagnosis and treatment of lung cancer. Her role will be to select eligible patients for [18F]FDG PET/CT screening and to collaborate with nuclear medicine physicians and oncologists in the management of patients following screening procedures. Roberto Ferrara (UO1) is a clinical oncologist at IRCCS Ospedale San Raffaele and a researcher at Università Vita-Salute San Raffaele in Milan. He is conducting several research projects in patients with lung cancer, particularly in the field of immunotherapy. His role will be to collaborate with nuclear medicine physicians and thoracic surgeons in the management of patients following screening procedures. Renato Cuocolo (UO2) is a radiologist at Azienda Ospedaliera Universitaria San Giovanni di Dio e Ruggi d'Aragona and Associate Professor at Università di Salerno. His research interests include the application of machine learning techniques to medical imaging and advanced image analysis in various oncological fields. His role will be to coordinate the activities of UO2 and to collaborate with the other UOs involved in the project. Andrea Ponsiglione (UO2) is a radiologist and post-doctoral research fellow. Teresa Perillo and Marco De Giorgi (UO2) are radiologist. Their role will be to evaluate and interpret LDCT screening images of patients enrolled in UO2. Paolo Soda (UO3) is Full Professor of Computer Science and Computer Engineering at Università Campus Bio-Medico di Roma and visiting Professor in AI and Biomedical Engineering at Umea University. His scientific interests include biomedical applications of classification techniques, signal, image and video processing and analysis. His role will be to coordinate the activities of UO3 and to collaborate with the other UOs involved in the project. Valerio Guarrasi (UO3) is a post-doctoral research fellow at Università Campus Bio-Medico di Roma. His role will be to develop an AI model to predict lung cancer risk based on clinical and imaging data in collaboration with UO4. Pierangelo Veltri (UO4) is Full Professor of Electronic and Computer Bioengineering at Università della Calabria. His research focuses on designing and application of deep learning techniques to the mining of electronic health records and biomedical data and the use of computational techniques for advanced analysis of radiology and nuclear medicine DICOM images. His role will be to coordinate the activities of UO4 and to collaborate with the other UOs involved in the project. Irina Trubitsyna (UO4) is a RTD-B at Università della Calabria. Her role will be to develop an AI model to predict lung cancer risk based on clinical and imaging data in collaboration with UO3.

5.13 Translational relevance and impact for the national health system (SSN)

What is already know about this topic?

Randomised controlled trials have demonstrated the performance of LDCT in detecting lung cancer (LC) in subjects with a history of smoking resulting in reduced mortality (de Koning et al, 2020). The added value of [18F]FDG PET have been less explored. Veronesi et al. investigated the efficacy of the selective use of PET/CT for indeterminate lung nodules in screening and showed a high sensitivity for the differential diagnosis of indeterminate nodules detected at baseline, nodules>15 mm and solid nodules (Veronesi et al 2015). AI models were used to optimise the selection of the target

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population for LDCT screening. A neural network prediction model was developed using clinical data from medical records, yielding an AUC of 0.87 in subjects aged >55 years (Chia-Han Yeh et al, 2021). Deep learning approaches are emerging as powerful tools for characterising lung nodules on CT and PET/CT images, achieving excellent performance (>94% AUC) (Ardila D. et al, 2019; Guo H. et al, 2022).

Details on what is already know about this topic

The NLST trial showed that LDCT screening had a higher positive rate than chest X-ray (at T0 27.3 vs 9.2%) but more than 94% of these were false positives, and reduced mortality by 20% (The NLST Research Team, 2011). The NELSON trial showed that LDCT screening was positive in 2.1% of cases, with a false-positive rate of 1.2%, and reduced mortality by 6% (de Koning et al, 2020). Veronesi et al. demonstrated that PET/CT has a specificity of 95% in nodules <10 mm detected by LDCT screening at baseline (Veronesi et al, 2015). Ardila et al. proposed a deep learning algorithm for lung cancer risk prediction using nodule volumetry on LDCT, achieving an AUC of 94.4% and outperforming radiologists with an absolute reduction of 11% in false positives and 5% in false negatives (Ardila D. et al, 2019). Guo et al. proposed a similar approach applied to PET images and achieved AUC values of 94.1% for lesions<10mm, 99.3% for lesions of 10-20mm and 99.8% for lesions>20 mm (Guo H. et al, 2022).

What this reasearch adds?

This project, stemming from the proven success of screening programmes in oncology, will bridge the still-existing gap in lung cancer screening and answer the strong unmet need of early diagnosis. Advancements in machine technology and availability of potent AI tools represent an unparalleled revolution in medical sciences, and this research programme aims to exploit both these exceptional tools to revolutionise the early management of this highly impactful and prevalent disease. Recent technological advancements in CT and PET/CT scanners will allow to perform exams with unprecedented accuracy and low administered dose to patients, reducing radiation exposure and healthcare costs. Moreover, this comprehensive multimodal approach will exploit AI to further enhance the diagnostic accuracy of screening examinations and personalised imaging schedule for each individual patient, potentially improving early diagnosis and streamlining screening programmes.

Details on what this reasearch adds

[18F]FDG PET/CT has only been used in lung cancer screening to characterise suspicious findings identified by LDCT (Pastorino U et al, 2003; Veronesi et al, 2015). This project will be the first to evaluate and clarify the potential contribution of [18F]FDG PET/CT to early diagnosis in lung cancer screening in a well-selected population of very high-risk patients. Smoking is a carcinogen that also significantly increases the risk of head and neck, gastrointestinal and bladder cancers (Weber MF, 2021). Therefore, high-risk individuals would benefit from a more comprehensive assessment to detect non-lung cancers that would be missed by chest LDCT (Chang ET et al, 2023). In addition, we will fill the gap of LDCT screening by using advanced artificial intelligence techniques to tailor screening programmes on a patient-by-patient basis in terms of population selection, screening interval, duration, and risk stratification for further imaging/diagnostic modalities (Novellis P et al, 2021).

What are the implications for public health, clinical practice, patient care?

Individualised planning of the intervals and duration of screening examinations will result in better patient compliance and selection of the population at risk, reduced radiation exposure from repeated LDCT scans over time to assess nodule growth, and a reduction in the number of invasive procedures not needed for characterisation of suspicious nodules. Patients with early-stage tumours will be identified and treated using minimally invasive techniques, reducing healthcare costs, morbidity and mortality associated with the disease.

Details on what are the implications for public health, clinical practice, patient care

The latest technological advances in PET/CT scanners allow a reduction in acquisition time and injected radiopharmaceutical activity while maintaining high image quality (Aide N et al, 2019). PET/CT will allow whole-body assessment with high levels of accuracy, outperforming LDCT in terms of specificity (95 vs 85%) (Guo L et al, 2023; Veronesi G et al, 2015). Moreover, PET/CT will detect other smoking-related cancers in a single examination. This

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approach could potentially reduce the rate of false-positive results, overdiagnosis, the amount of unnecessary additional diagnostic and invasive testing, and therefore costs. Whole-body low dose PET/CT has the potential to reduce morbidity and mortality in high-risk individuals in the near future.

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6 - Budget

Total proposed budget (Euro)				
Costs	TOTAL BUDGET	Co-Funding	List of costs proposed for funding to the MOH	Percentage of total proposed to the MOH
1 Staff Salary	0,00	-0,00	not permitted	0,00
2 Researchers' Contracts	600.000,00	0,00	600.000,00	60,00
3a.1 Equipment (Leasing -	0,00	0,00	0,00	0,00
3a.2 Equipment (buying)	0,00	0,00	0,00	0,00
3b Supplies	0,00	0,00	0,00	0,00
3c Model Costs	0,00	0,00	0,00	0,00
4 Subcontracts *	75.000,00	0,00	75.000,00	7,50
5 Patient Costs	111.814,00	0,00	111.814,00	11,18
6 IT Services and Data Bases	15.000,00	0,00	15.000,00	1,50
7 Travels	20.000,00	0,00	20.000,00	2,00
8 Publication Costs	30.000,00	0,00	30.000,00	3,00
9 Dissemination	20.000,00	0,00	20.000,00	2,00
10 Overheads *	60.186,00	0,00	60.186,00	6,02
11 Coordination Costs	68.000,00	0,00	68.000,00	6,80
Total	1.000.000,00	0,00	1.000.000,00	100,00

* percentage calculated as average value between all the Operating Units.

Report the Co-Funding Contributor:

NA

Budget Justification	
1 Staff Salary	NA
2 Researchers' Contracts	One researcher contract at UO1 for 2 years; three under-40 researchers at UO2 for 2 years; one researcher contract at UO3 for 2 years and one researcher contract at UO3 for 1 year; two researcher contract at UO1 for 2 years.
3a.1 Equipment (Leasing - Rent)	NA
3a.2 Equipment (buying)	NA
3b Supplies	NA



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3c Model Costs	NA
4 Subcontracts	CRO
5 Patient Costs	Radiopharmaceutical production costs ([18F]FDG); patients insurance policy.
6 IT Services and Data Bases	Infrastructure for data sharing and software; website, interfaces and computational tools.
7 Travels	Travels for participation to international conferences and dissemination
8 Publication Costs	Publication in high-impact journals
9 Dissemination	Participation for presentation of the project results at international congresses
10 Overheads	overheads
11 Coordination Costs	Data manager contract for 2 years

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Proposed total budget UO1 Institution: Ospedale San Raffaele - Milano (Euro)

Costs	TOTAL BUDGET	Co-Funding	List of costs proposed for funding to the MOH	Percentage of total proposed to the MOH
1 Staff Salary	0,00	-0,00	not permitted	0,00
2 Researchers' Contracts	80.000,00	0,00	80.000,00	21,25
3a.1 Equipment (Leasing - Rent)	0,00	0,00	0,00	0,00
3a.2 Equipment (buying)	0,00	0,00	0,00	0,00
3b Supplies	0,00	0,00	0,00	0,00
3c Model Costs	0,00	0,00	0,00	0,00
4 Subcontracts	75.000,00	0,00	75.000,00	19,92
5 Patient Costs	111.814,00	0,00	111.814,00	29,70
6 IT Services and Data Bases	0,00	0,00	0,00	0,00
7 Travels	6.500,00	0,00	6.500,00	1,73
8 Publication Costs	10.500,00	0,00	10.500,00	2,79
9 Dissemination	5.000,00	0,00	5.000,00	1,33
10 Overheads	19.726,00	0,00	19.726,00	5,24
11 Coordination Costs	68.000,00	0,00	68.000,00	18,06
Total	376.540,00	0,00	376.540,00	100,00



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Budget Justification

1 Staff Salary	NA
2 Researchers' Contracts	One researcher contract for 2 years
3a.1 Equipment (Leasing - Rent)	NA
3a.2 Equipment (buying)	NA
3b Supplies	NA
3c Model Costs	NA
4 Subcontracts	CRO
5 Patient Costs	Radiopharmaceutical production costs ([18F]FDG); patients insurance policy.
6 IT Services and Data Bases	NA
7 Travels	Travels for participation to international conferences and dissemination
8 Publication Costs	Publication in high-impact journals
9 Dissemination	Participation for presentation of the project results at international congresses
10 Overheads	Overheads
11 Coordination Costs	Data manager contract for 2 years

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Proposed total budget UO2 Institution: Ospedale San Giovanni di Dio e Ruggi d'Aragona (Euro)

Costs	TOTAL BUDGET	Co-Funding	List of costs proposed for funding to the MOH	Percentage of total proposed to the MOH
1 Staff Salary	0,00	-0,00	not permitted	0,00
2 Researchers' Contracts	240.000,00	0,00	240.000,00	87,62
3a.1 Equipment (Leasing - Rent)	0,00	0,00	0,00	0,00
3a.2 Equipment (buying)	0,00	0,00	0,00	0,00
3b Supplies	0,00	0,00	0,00	0,00
3c Model Costs	0,00	0,00	0,00	0,00
4 Subcontracts	0,00	0,00	0,00	0,00
5 Patient Costs	0,00	0,00	0,00	0,00
6 IT Services and Data Bases	0,00	0,00	0,00	0,00
7 Travels	4.500,00	0,00	4.500,00	1,64
8 Publication Costs	6.500,00	0,00	6.500,00	2,37
9 Dissemination	5.000,00	0,00	5.000,00	1,83
10 Overheads	17.920,00	0,00	17.920,00	6,54
11 Coordination Costs	not permitted	not permitted	not permitted	0,00
Total	273.920,00	0,00	273.920,00	100,00

 <i>Ministero della Salute</i> Direzione generale della ricerca e dell'innovazione in sanità PNRR: M6/C2_CALL 2023 Full Proposal	 Finanziato dall'Unione europea NextGenerationEU
Project Code: PNRR-MCNT2-2023-12377755	Call section: Malattie Croniche non Trasmissibili (MCnT2) ad alto impatto sui sistemi sanitari
Applicant Institution: Ospedale San Raffaele - Milano	Applicant/PI Coordinator: CHITI ARTURO

Budget Justification	
1 Staff Salary	NA
2 Researchers' Contracts	three under-40 researchers for 2 years
3a.1 Equipment (Leasing - Rent)	NA
3a.2 Equipment (buying)	NA
3b Supplies	NA
3c Model Costs	NA
4 Subcontracts	NA
5 Patient Costs	NA
6 IT Services and Data Bases	NA
7 Travels	Travels for participation to international conferences and dissemination
8 Publication Costs	Publication in high-impact journals
9 Dissemination	Participation for presentation of the project results at international congresses
10 Overheads	Overheads
11 Coordination Costs	NA

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Proposed total budget UO3 Institution: Campus Biomedico - Roma (Euro)

Costs	TOTAL BUDGET	Co-Funding	List of costs proposed for funding to the MOH	Percentage of total proposed to the MOH
1 Staff Salary	0,00	-0,00	not permitted	0,00
2 Researchers' Contracts	120.000,00	0,00	120.000,00	76,81
3a.1 Equipment (Leasing - Rent)	0,00	0,00	0,00	0,00
3a.2 Equipment (buying)	0,00	0,00	0,00	0,00
3b Supplies	0,00	0,00	0,00	0,00
3c Model Costs	0,00	0,00	0,00	0,00
4 Subcontracts	0,00	0,00	0,00	0,00
5 Patient Costs	0,00	0,00	0,00	0,00
6 IT Services and Data Bases	10.000,00	0,00	10.000,00	6,40
7 Travels	4.500,00	0,00	4.500,00	2,88
8 Publication Costs	6.500,00	0,00	6.500,00	4,16
9 Dissemination	5.000,00	0,00	5.000,00	3,20
10 Overheads	10.220,00	0,00	10.220,00	6,54
11 Coordination Costs	not permitted	not permitted	not permitted	0,00
Total	156.220,00	0,00	156.220,00	100,00

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Budget Justification	
1 Staff Salary	NA
2 Researchers' Contracts	one researcher contract for 2 years and one researcher contract for 1 year
3a.1 Equipment (Leasing - Rent)	NA
3a.2 Equipment (buying)	NA
3b Supplies	NA
3c Model Costs	NA
4 Subcontracts	NA
5 Patient Costs	NA
6 IT Services and Data Bases	Infrastructure for data sharing
7 Travels	Travels for participation to international conferences and dissemination
8 Publication Costs	Publication in high-impact journals
9 Dissemination	Participation for presentation of the project results at international congresses
10 Overheads	Overheads
11 Coordination Costs	NA

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Proposed total budget UO4 Institution: DIMES Università della Calabria (Euro)

Costs	TOTAL BUDGET	Co-Funding	List of costs proposed for funding to the MOH	Percentage of total proposed to the MOH
1 Staff Salary	0,00	-0,00	not permitted	0,00
2 Researchers' Contracts	160.000,00	0,00	160.000,00	82,76
3a.1 Equipment (Leasing - Rent)	0,00	0,00	0,00	0,00
3a.2 Equipment (buying)	0,00	0,00	0,00	0,00
3b Supplies	0,00	0,00	0,00	0,00
3c Model Costs	0,00	0,00	0,00	0,00
4 Subcontracts	0,00	0,00	0,00	0,00
5 Patient Costs	0,00	0,00	0,00	0,00
6 IT Services and Data Bases	5.000,00	0,00	5.000,00	2,59
7 Travels	4.500,00	0,00	4.500,00	2,33
8 Publication Costs	6.500,00	0,00	6.500,00	3,36
9 Dissemination	5.000,00	0,00	5.000,00	2,59
10 Overheads	12.320,00	0,00	12.320,00	6,37
11 Coordination Costs	not permitted	not permitted	not permitted	0,00
Total	193.320,00	0,00	193.320,00	100,00

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Budget Justification	
1 Staff Salary	NA
2 Researchers' Contracts	two researcher contract for 2 years
3a.1 Equipment (Leasing - Rent)	NA
3a.2 Equipment (buying)	NA
3b Supplies	NA
3c Model Costs	NA
4 Subcontracts	NA
5 Patient Costs	NA
6 IT Services and Data Bases	Website, interfaces and computational tools.
7 Travels	Travels for participation to international conferences and dissemination
8 Publication Costs	Publication in high-impact journals
9 Dissemination	Participation for presentation of the project results at international congresses
10 Overheads	Overheads
11 Coordination Costs	NA



Ministero della Salute

Direzione generale della ricerca e dell'innovazione in sanità

PNRR: M6/C2_CALL 2023 Full Proposal



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Applicant Institution: Ospedale San Raffaele - Milano	Applicant/PI Coordinator: CHITI ARTURO

Principal Investigator Data

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Nome: ARTURO
Genere: M
Codice fiscale: CHTRTR62E17A944W
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Qualifica: Medico
Struttura: Medicina Nucleare
Istituzione: IRCCS Istituto Clinico Humanitas
Datore/ente di lavoro? Yes
Datore/ente di lavoro SSN? No
Nome datore/ente di lavoro non SSN: Università Vita-Salute San Raffaele
Nome istituzione SSN: IRCCS San Raffaele
Tipo contratto: Lavoro Subordinato a Tempo Indeterminato

Con l'invio della presente proposta si dichiara che la stessa o parti significative di essa non sono oggetto di altri finanziamenti pubblici o privati e che di conseguenza vi è assenza del c.d. doppio finanziamento ai sensi dell'art. 9 del Regolamento (UE) 2021/241, ossia che non ci sia una duplicazione del finanziamento degli stessi costi da parte di altri programmi dell'Unione, nonché con risorse ordinarie da Bilancio statale.

By submitting this proposal, I declare that no significant part or parts of it are recipient of any other public or private funding and that consequently there isn't any so-called double financing pursuant to art. 9 of Regulation (EU) 2021/241, i.e. that there is no duplication in the financing of the same costs by other European Union programs or any other ordinary resources from the State budget.

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Project validation result
